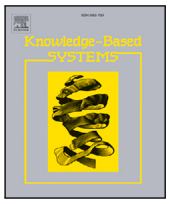
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NeuroSymbolic-Chain: A neuro-symbolic framework for intelligent disaster chain analysis from multi-source text

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ABSTRACT

Under non-stationary climate conditions, meteorological disasters have evolved into complex cross-system cascading networks whose catastrophic impacts frequently stem from latent secondary disaster chains. Despite strong semantic comprehension, Large Language Models (LLMs) remain susceptible to logical hallucinations during long-chain reasoning because existing generative paradigms do not embed rigorous physical constraints in probabilistic text generation. To address this, we introduce NeuroSymbolic-Chain, a neuro-symbolic framework synergizing LLMs with Dynamic Causal Knowledge Graphs. This framework orchestrates a closed-loop neuro-symbolic iteration: LLM-Driven Knowledge Graph Construction employs Ontology-Guided Prompt Engineering (OGPE) to extract entities from heterogeneous text streams and construct and incrementally update a DCKG; Graph-Based Propagation Reasoning utilizes Topology-Aware GraphRAG and subgraph isomorphism to deduce implicit risk trajectories via historical motifs; and Graph-Guided Context Engineering formalizes immutable physical consistency rules into strict logical constraints, injecting them into the generative process to mitigate hallucinations at their source. To rigorously substantiate the methodological novelty of this closed-loop architecture, we conduct comprehensive mechanism-isolation and same-backbone analyses. Experimental results show that, compared with state-of-the-art baselines, NeuroSymbolic-Chain improves Chain Accuracy by 5.9% and reduces the Logic Violation Rate by 52.4%. The case study of the July 2023 extreme rainstorm in Beijing shows that the framework can identify structural risks (e.g., subgrade hollowing) missed by conventional models. These results support interpretable disaster-chain analysis for resilient urban planning.

1. Introduction

In July 2023, Beijing experienced an extreme rainstorm of historic magnitude [1], hereafter referred to as the July 2023 extreme rainstorm in Beijing. The event triggered severe urban waterlogging, infrastructure collapse, and substantial economic loss. Post-disaster forensics revealed that many catastrophic outcomes were not isolated incidents but products of an underlying cascading sequence: Continuous Heavy Rainfall → River Level Surge → Lateral Erosion → Riparian Subgrade Hollowing and Collapse. Conventional single-dimension monitoring systems rarely capture such multi-source, cross-system event chains. This limitation highlights a critical gap in meteorological disaster analysis: the lack of methods that can parse latent evolutionary logic from voluminous, fragmented text streams and predict cross-domain cascading effects.

Recurrent in historical records — from rainstorm-induced traffic paralysis to drought-driven power shortages — these cascading patterns indicate that modern urban disasters have evolved into complex socio-natural risk networks. Crucially, their evolutionary mechanisms remain obscured by the semantic gap created by noisy, asynchronous reports from social media and news streams [2,3]. This challenge has made the intelligent parsing of meteorological disaster chains a focal problem in disaster knowledge engineering [4].

While early research leveraged rule-based or traditional Natural Language Processing (NLP) for entity-level extraction, these methods, though efficient at identifying discrete entities, fail to capture underlying mechanistic dynamics. Deep learning and Knowledge Graph (KG) techniques later introduced dynamic representations that partially addressed temporal forecasting [5]. More recently, Retrieval-Augmented

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Generation (RAG) methods built on large language models, including GraphRAG [6] and TimeR4 [7] within the broader RAG paradigm [8], have extended this line toward more complex temporal reasoning.

Broader machine learning paradigms have also been effective in disaster and hydrometeorological modeling. Neural nowcasting models can capture the nonlinear evolution of precipitation while quantifying predictive uncertainty [9]. Hybrid sequence-GP frameworks improve reservoir-level prediction under uncertainty [10]. Topology-aware flood forecasting extends warning lead times [11], while ensemble learning improves robustness and interpretability in landslide susceptibility mapping [12]. Consequently, purely statistical and data-driven methods remain highly effective, offering distinct advantages in forecasting, susceptibility assessment, and the quantification of uncertainty.

Despite these advances, existing methods face three fundamental limitations when parsing complex disaster chains from heterogeneous text.

1. **Semantic Gap Challenge:** This limitation stems from the inherent disparities among diverse data sources. When the objective is to reconstruct latent causal chains from text rather than merely forecasting numerical targets, current approaches lack the capacity to effectively integrate structured monitoring data with unstructured social text. Traditional NLP typically captures element-level correlation but fails to infer rigorous causality, such as the specific temporal-causal sequence where rainfall drives subgrade hollowing.
2. **Logical Hallucination:** Absent strict external constraints, general-purpose LLMs are highly susceptible to logical hallucinations during long-chain reasoning [13]. These failures manifest as temporal inversion (e.g., floods causing rainstorms), causal over-inference, and structurally invalid chain composition.
3. **Domain-Specific Constraints:** Current graph reasoning and RAG techniques prioritize global retrieval or simplistic linear temporal decay, lacking domain-specific causal constraints. Relying heavily on semantic similarity, GraphRAG fails to capture the latent topological isomorphism of disaster chains, leaving it ill-equipped to model long-latency cumulative effects or resolve logical cycles.

These limitations stem from a profound challenge: how to embed rigorous physical-logic cross-validation within purely semantic generative processes. Without resolving this, even state-of-the-art models cannot provide the high-confidence decision support required by urban managers.

To address these challenges, we introduce **NeuroSymbolic-Chain**, a neuro-symbolic framework synergizing LLMs with Dynamic Causal Knowledge Graphs (DCKG). Guided by the Principles of Narrative Invariance (PNI) — a set of axioms ensuring that generative narratives strictly adhere to physical laws such as temporal sequence and causal transitivity — this framework shifts the emphasis from probabilistic generation toward logically constrained causal reasoning. Specifically, NeuroSymbolic-Chain comprises three tightly integrated components that form a closed-loop neuro-symbolic system:

1. LLM-Driven Knowledge Graph Construction. Addressing data heterogeneity, this module processes heterogeneous multi-source textual evidence streams (e.g., social media and government bulletins) via an LLM-based neuro-symbolic extractor. Utilizing Ontology-Guided Prompt Engineering (OGPE) and a dual-stream textual state strategy, it constructs and incrementally updates a DCKG as new evidence arrives, with source timestamps normalized to an hourly resolution, thereby capturing disaster entities and their evolutionary states. The specific contribution of ontology-guided extraction to semantic normalization is validated through ablation studies, which isolate its impact on entity stability.

2. Graph-Based Propagation Reasoning. Focused on implicit risk discovery, this module goes beyond surface-level semantics by employing Topology-Aware GraphRAG. It performs subgraph isomorphism matching and path reasoning within the graph. By incorporating historical motifs, the module deduces secondary disaster chains that lack explicit textual representation but remain structurally plausible under the established constraint set. The methodological superiority of this topology-aware mechanism over standard graph retrieval is demonstrated across various LLM backbones (e.g., Qwen3, GPT-4o) to eliminate the confounding effects of model scale.

3. Graph-Guided Context Engineering. Designed to reduce logically inconsistent generations, this module utilizes DFS-based topological serialization to transform inferred risk subgraphs into structured prompts. By injecting PNI constraints — temporal axioms, causal transitivity, and semantic intensity homomorphism — it guides the LLM to generate analysis reports within strict physical and logical boundaries. We validate the effectiveness of this constraint-guided generation by analyzing the granular decomposition of logical violations, thereby demonstrating that the proposed mechanism fundamentally mitigates physical inconsistencies rather than merely altering the distribution of error types.

For terminological precision, Table 1 defines the core terminology used throughout this paper.

The remainder of this paper is organized as follows. Section 2 reviews the related literature. Sections 3–5 detail ontology-guided graph construction, graph-based propagation reasoning, and graph-guided context engineering, respectively. Section 6 outlines the experimental settings and reports the quantitative results. Section 7 presents the Beijing case study. Section 8 discusses the limitations, ethical considerations, and future directions. Appendix A provides the core artifacts for reproducibility, including the configuration, ontology design, prompt templates, and the statement of data and code availability.

2. Related work

2.1. Neuro-symbolic AI for knowledge graphs

Neuro-symbolic AI, which integrates the perceptual capabilities of neural networks with the logical rigor of symbolic systems, has emerged as a critical approach to mitigate unconstrained LLM generation. Recent studies constrain LLMs at several levels rather than through a single mechanism. Schema-level output control improves structural compliance [14]. Ontology-guided reasoning injects domain relations into multi-hop inference [15,16]. Linearized KG evidence can improve fact-intensive prompting compared with free-form reformulations [17]. Question-specific subgraph retrieval further reduces irrelevant knowledge during answer generation [18]. In domain KG construction, ontology-grounded few-shot CoT prompting improves extraction quality from technical text [19]. Retrieve-then-revise pipelines strengthen factual correction through external verification [20,21], while logic-constrained decoding narrows lexical realization under symbolic rules [22]. These studies show that explicit structure and symbolic feedback can stabilize generation. However, they mainly regulate output format, local factuality, or generic reasoning structure, whereas our setting requires direct control over disaster-specific physical regularities such as temporal order, acyclic propagation, and intensity compatibility in continuously updated observational text.

Science-oriented LLM systems have further shifted the emphasis from free-form generation to evidence-intensive reasoning. Tool-augmented agents support scientific problem solving [23]. Retrieval-grounded literature agents improve citation-backed synthesis [24,25], and multi-agent critique frameworks strengthen iterative refinement [26]. The connection to this study is methodological rather than domain-specific: these systems highlight the value of explicit intermediate structure and feedback, but they do not operate under the physical and cross-source consistency constraints required for disaster-chain reconstruction. In contrast to existing neuro-symbolic methods that primarily address generic logical constraints or local factuality, the proposed framework distinctively integrates meteorology-specific physical rules — such as temporal ordering, acyclic propagation, and intensity compatibility — into an executable graph write-back loop for processing heterogeneous disaster texts.

Table 1
Core terminology.

Term	Definition in this study	Representative indicator
Hallucination taxonomy		
Logical hallucination	Structural or physical inconsistency in a generated or inferred disaster chain; this is the primary hallucination type targeted by NeuroSymbolic-Chain.	Temporal inversion, causal cycles, invalid propagation directions, and elevated logic-violation rate (LVR).
Narrative hallucination	Unsupported or factually inaccurate disaster narratives relative to the available evidence, distinct from logical inconsistency.	Unsupported event claims and lower Fact Consistency.
Semantic hallucination	Semantic drift in event meaning or severity despite fluent surface form.	Overstated or understated severity and incompatible state descriptions.
Framework concepts		
Dynamic Causal Knowledge Graph (DCKG)	A time-indexed graph whose node states and edges evolve as new textual evidence is incorporated.	State transitions and edge additions triggered by incoming observations.
Dynamic update	An online incremental update mechanism rather than offline batch reconstruction, with timestamps normalized to hourly granularity.	Event-driven write-back, Bayesian conflict resolution, and time-aware forgetting.

2.2. Graph-based reasoning for disaster-chain analysis

Graph-based reasoning has become ubiquitous for modeling non-linear system dynamics, particularly as Retrieval-Augmented Generation (RAG) evolves from vector-based retrieval [27] to GraphRAG [28]. Recent research improves structured retrieval along three main directions. Knowledge graph-guided retrieval organizes evidence beyond surface lexical overlap and supports cross-document reasoning [29,30]. Temporal retrieval methods introduce chronological relevance into the retrieval stage [31]. Other systems strengthen structure through hierarchical abstraction and memory-based graph recall [32,33], or through graph indexing and dynamic relation retrieval [34,35]. Related efforts also convert text into graph-structured intermediates for multi-step reasoning [36] and expose the brittleness of long-context multi-hop inference when crucial links are implicit [37]. MuRAG extends this line to multimodal evidence integration [38].

Despite these advancements, existing graph reasoning methods remain inadequate for in-depth disaster attribution. First, most retrieval pipelines continue to prioritize semantic proximity over topological isomorphism, although geographically distinct disaster events frequently share structurally similar propagation subgraphs. Second, existing time-aware retrieval mechanisms are typically designed for temporal relevance ranking rather than causal validity verification. Consequently, they are ill-equipped to identify reverse-temporal links, causal cycles, or energy-incompatible propagation transitions. These studies motivate our graph reasoning module, but they largely optimize retrieval adequacy. By contrast, NeuroSymbolic-Chain treats retrieval as candidate generation and then applies motif matching plus physical-rule filtering to enforce causal validity. Although recent GraphRAG variants prioritize dual-level retrieval or temporal traversal, the methodological distinction of the proposed framework lies in integrating topology-aware motif retrieval with deterministic physical-rule validation and Bayesian conflict arbitration. This integration proactively eliminates logically invalid candidate paths rather than merely broadening the retrieval scope.

2.3. Disaster chain modeling

In contemporary disaster management, disasters have evolved into complex socio-natural risk networks — manifested in real time through voluminous multi-source text streams — here referred to as cascading disaster chains [39]. Existing work on this problem largely approaches cascading effects from either dynamic graph evolution or streaming logical state management. Dynamic graph representations help track evolving disaster interactions over time [5], whereas stream-reasoning frameworks emphasize robust state maintenance under continuously arriving evidence [40].

In the modeling of complex non-linear disasters, generalized machine learning paradigms have also proved effective. For instance, neural networks [41,42] and ensemble methods [43,44] can effectively capture spatiotemporal dependencies and enhance predictive robustness; meanwhile, Gaussian process regression [45,46] and graphical models [47,48] exhibit strong performance in quantifying uncertainties and simulating complex patterns.

While these studies establish a foundational understanding of cascading effects, current modeling approaches face three fundamental limitations. First, existing dynamic graph methods predominantly focus on numerical prediction across discrete time steps, often neglecting the non-linear temporal decay inherent in the informational value of unstructured text. Second, traditional models struggle to bridge the semantic gap between heterogeneous sources, failing to effectively synthesize structured monitoring data with unstructured social discourse to infer deep logical chains. Third, the absence of dynamic resolution mechanisms for conflicting multi-source information leaves systems vulnerable to "consensus hallucinations" when processing contradictory disaster reports.

2.4. Causal extraction from text

Causal extraction from text represents a longstanding challenge in Natural Language Processing (NLP). The recent shift toward generative models has moved the field from pipeline-based relation mining to generation-centric reasoning. Recent agent-based systems improve robustness for complex information extraction [49], yet general-purpose LLMs remain vulnerable to temporal inversion and causal over-inference in long-chain generation [50]. Causal de-confounding has also been shown to improve the detection of erroneous but structurally plausible triples in KG error detection [51]. The key limitation is not only pair-level extraction accuracy, but also the absence of mechanisms that preserve chain-level temporal direction, physical plausibility, and conflict resolution across sources. The most relevant neighboring literature falls into three streams. Structural causal models and do-calculus formalize interventional reasoning over explicit causal graphs [52]. Granger-style approaches applied to text operationalize causality through predictive precedence in temporally aligned signals [53,54]. Temporal KG research, in turn, emphasizes time validity and evolving causal subhistories during reasoning [55,56]. These streams clarify the conceptual boundary of our study: NeuroSymbolic-Chain operates on observational multi-source text, using topological motifs as structural candidates and physical rules as rigid filters to reject purely statistical correlations. It is crucial to clarify that the proposed framework does not attempt to conduct strict causal identification from purely observational text. Instead, it generates and structurally validates physically plausible candidate chains to facilitate text-based emergency decision-making, thereby serving as a complement to, rather than a substitute for, physical simulations.

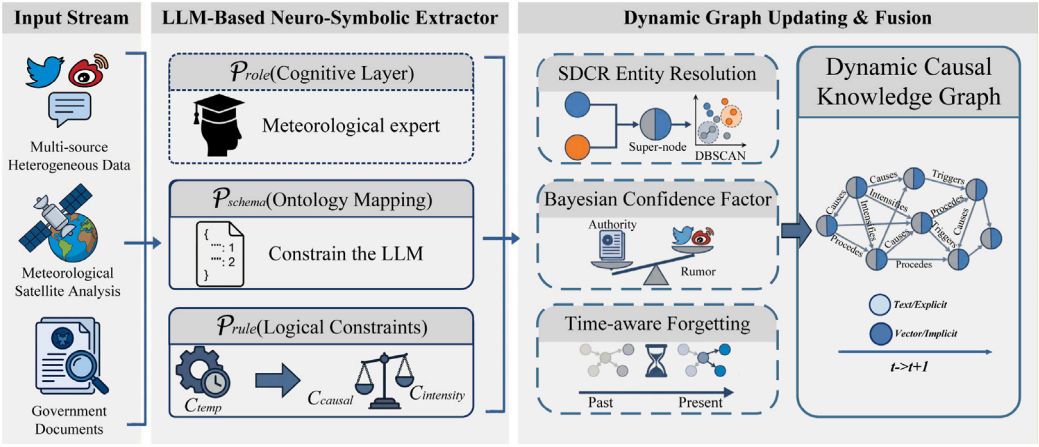


Fig. 1. DCKG construction. OGPE-based extraction is paired with entity disambiguation, Bayesian conflict resolution, and time-aware forgetting to support incremental graph updates.

3. LLM-driven knowledge graph construction

This section details the construction of a Dynamic Causal Knowledge Graph (DCKG) derived from heterogeneous multi-source textual evidence streams via Large Language Models (LLMs), where causal edges encode directed cascading hypotheses supported by physical propagation axioms, heterogeneous textual evidence, and consistency constraints, rather than strict causal effects identified through interventions. Throughout this paper, "dynamic" has two linked meanings: the DCKG stores time-indexed states that evolve as evidence accumulates, whereas the update mechanism performs online incremental write-back rather than offline batch reconstruction, with source timestamps normalized to an hour granularity. Because meteorological disaster data is inherently noisy, unstructured, and exhibits a long-tail distribution, conventional entity extraction fails to capture underlying causal mechanisms. To address this, we introduce an Ontology-Guided Prompt Engineering (OGPE) mechanism that integrates a Dual-Stream Textual State with Bayesian conflict resolution. This approach ensures precise graph construction and robust dynamic evolution, as illustrated in Fig. 1.

3.1. Ontology-guided extraction

We process a heterogeneous multi-source textual evidence stream $D = \{d_1, d_2, \ldots, d_n\}$ — encompassing social media feeds, government bulletins, and technical interpretation reports — using an LLM-based neuro-symbolic extractor to identify disaster entities and their causal dependencies.

Formal Definition & Dual-Stream Textual Representation. The Dynamic Causal Knowledge Graph (DCKG) is formalized as $G = (V, E, \tau_v, \tau_e, t)$, where V represents the set of event nodes and $E \subseteq V \times V$ denotes directed causal edges. Here, τ_v and τ_e map entities and relations to mutually exclusive ontological spaces, while $t : V \to \mathbb{R}^+$ serves as an absolute temporal anchor. Edges labeled as "causes" denote directed cascading hypotheses supported by heterogeneous multi-source textual evidence, physical propagation axioms, and domain constraints.

To bridge the semantic gap between subjective sentiment and objective fact in unstructured text, we employ a dual-stream textual representation strategy. The state of an arbitrary node v is defined as a hybrid feature vector h_v, constructed by concatenating projections from two complementary textual evidence views:

1. **The Explicit Narrative Stream** (h_{exp})**:** Utilizes a mapping function f_{grade} to quantify discrete modifiers (e.g., "Red Warning") into ordinal vectors.
2. **The Implicit Latent Stream** (h_{imp})**:** A high-dimensional dense vector ($h_{imp} \in \mathbb{R}^d$) designed to capture textual urgency and emotional intensity.

$$S_v = \Psi(v) = \frac{h_{imp}(v) \cdot w_{severe}}{\|h_{imp}(v)\| \cdot \|w_{severe}\|} \tag{1}$$

To calibrate these representations, we introduce a composite loss function $L_{intensity} = \lambda_1 L_{MSE} + \lambda_2 L_{rank}$. This function anchors predictions to absolute physical reference levels via Mean Squared Error (L_{MSE}) while preserving relative intensity hierarchies (e.g., Rainstorm $\prec$ Torrential Rain) via Ranking Loss (L_{rank}), effectively mitigating the tendency of the LLM toward hyperbolic or understated generation.

Ontology-Guided Prompt Engineering (OGPE). To achieve high-precision extraction in zero-shot settings, we construct a task-specific disaster ontology, denoted as P_Ω, and instantiate it through a JSON-oriented runtime schema. This ontology ensures the fine-grained coverage of meteorological hazard cascades and aligns with the Chinese warning terminology required by the runtime interface, capabilities that existing public ontologies do not directly provide. Specifically, we combine top-down domain scoping derived from official hazard typologies and warning vocabularies with a bottom-up consolidation of high-frequency events and relation expressions extracted from historical disaster texts from multiple sources. This process is followed by a manual adjudication step, where experts in meteorological disasters and emergency management cross-check synonymy, scope overlaps, and class-boundary conflicts based on conservative criteria for physical plausibility and operational terminology. The resulting ontology is structured into upper domains, canonical classes, and normalized leaf classes, which are subsequently formalized in machine-readable Turtle/OWL formats and summarized in Appendix A.2. Instead of directly exposing the hierarchy of the ontology, the prompt tiers of OGPE serve as an execution interface via the runtime schema. Consequently, we decompose the extraction prompt into three interdependent logical tiers:

- **Cognitive Layer** (P_{role})**:** Primes the model as a meteorological disaster expert, activating domain-specific latent semantic spaces.
- **Ontology Mapping Layer** (P_{schema})**:** Explicitly translates node types τ_v and causal relations τ_e into JSON Schema descriptors, mandating a strict {Subject, Relation, Object, Strength} quadruple output format.
- **Logical Constraint Layer** (P_{rule})**:** This layer represents the formal implementation of PNI during the extraction phase. From the perspective of general constraint design, PNI adapts concepts of temporal ordering and safety constraints that have been extensively studied in temporal logic and planning systems [57].

The task-specific contribution of this work lies in instantiating these general concepts as disaster-chain predicates $C = \{C_{temp}, C_{causal}, C_{intensity}\}$ and coupling them with ontology-grounded extraction, deterministic symbolic validation, and graph write-back operations. These predicates are executed by an external deterministic symbolic validator before candidate quadruples are integrated into the graph:

- **Temporal Topology Constraint** (C_{temp})**:** This predicate evaluates each candidate causal pair (v_i, v_j) against a physical time-lag threshold δ_{min}, rejecting the edge if $t(v_i)+\delta_{min} > t(v_j)$.
- **Causal Transitivity Constraint** (C_{causal})**:** To explicitly prevent the formation of logical cycles, this constraint eliminates edges lacking geographical adjacency or physical connectivity.
- **Semantic Intensity Homomorphism** ($C_{intensity}$)**:** This final predicate ensures that the downstream disaster intensity $\lambda(v_j)$ is compatible with the upstream energy input, effectively removing candidates that violate the principles of energy conservation.

Under this setting, C_{causal} serves to constrain the plausibility of spatial propagation and prevent causal cycles, rather than to enforce strict assumptions of causal faithfulness. Through rigid non-textual discriminative mechanisms, C_{temp}, C_{causal}, and $C_{intensity}$ filter out a substantial number of purely statistical correlations. However, latent confounders and selection biases in observational texts may still persist due to variations in infrastructure quality, soil composition, sensor coverage, or reporting density [58]. Consequently, this set of rules provides necessary conditions for constraint-guided generation and consistency verification, rather than sufficient criteria for causal identification.

3.2. Dynamic graph updating

Unlike static counterparts, the DCKG evolves continuously over time t. We address the inherent fragmentation and contradiction of multi-source data through a tripartite mechanism: Semantic Density-based Cross-document Coreference Resolution (SDCR), streaming state updates, and Bayesian conflict resolution.

Semantic Density-based Cross-document Coreference Resolution (SDCR). To resolve fragmented entities dispersed across heterogeneous documents, we define a multi-view parameterized distance metric $D(v_i, v_j)$ to establish entity identity:

$$D(v_i, v_j) = \omega_1(1 - \cos(h^i_{imp}, h^j_{imp})) + \omega_2|t_i - t_j| + \omega_3 d_J(N(v_i), N(v_j)) \quad (2)$$

Here, the first term captures implicit semantic embedding similarity, the second accounts for temporal divergence, and the third (d_J) quantifies topological neighborhood dissimilarity via the Jaccard coefficient. Leveraging this metric, the DBSCAN algorithm collapses discrete entities [59] within high-density semantic clusters into unified super-nodes. This process effectively reduces multi-source redundancy, distilling scattered descriptions into unique, distinct physical events.

State Quantization and Time-aware Forgetting. To model the non-linear decay of information value characteristic of physical systems, we introduce a Time-aware Forget Gate. Upon receiving a new observation e_{new} at time t, node states evolve according to the following dynamic update equation:

$$h_v(t) = \eta(\Delta t) \cdot h_v(t-1) + (1 - \eta(\Delta t)) \cdot Enc(e_{new}) \quad (3)$$

where $\eta(\Delta t) = \exp(-\lambda \cdot \Delta t)$ serves as an exponential decay function governed by the time interval. During preprocessing, multi-source timestamps are floored and normalized to an hour granularity; therefore, Δt is strictly measured in hours. This mechanism ensures the graph rapidly adapts to emerging disaster dynamics while actively suppressing interference from obsolete data. Rather than parameterizing time-varying continuous causal strengths at the edge level, as is common in temporal KG frameworks, the model applies temporal decay to node states. This discrete-event formulation aligns with the bursty and irregular reporting cadence of disaster text streams, thereby reducing the risk of overfitting sparse observations.

Bayesian Conflict Resolution. To resolve narrative conflicts inherent in heterogeneous data — such as discrepancies between official bulletins and social media rumors — we construct a Bayesian Confidence Factor (BCF). The BCF functions as a weighting mechanism maximizing posterior probability, integrating source authority (α_{source}), temporal decay (β), and cognitive uncertainty ($U(s)$).

$$BCF(s) = \left(\frac{1}{U(s) + \varepsilon}\right) \cdot \alpha_{source} \cdot e^{-\beta \cdot (t_{now} - t_s)} \quad (4)$$

where $\varepsilon = 10^{-5}$ is a fixed numerical smoothing constant, excluded from hyperparameter search and used specifically to prevent division-by-zero exceptions when $U(s) = 0$ under extreme evidence-sparse conditions. Graph updates strictly adhere to the Maximum A Posteriori (MAP) criterion: system state updates occur only when the BCF of new evidence significantly exceeds that of existing data. This ensures that the graph maintains alignment with a more reliable, evidence-weighted state estimate, even within a noisy semantic space.

4. Graph-based propagation reasoning

To mitigate the logical hallucinations inherent in LLM-driven long-chain reasoning, we introduce a dedicated neuro-symbolic graph reasoning module. This component transforms discrete events extracted from unstructured text into a structured Dynamic Event Graph (DEG). Specifically, DEG denotes the current event-level subgraph instantiated from the evolving DCKG, whereas DKG denotes the static template graph encoding physically valid patterns of disaster chains. By aligning and reasoning against a predefined static Disaster Knowledge Graph (DKG), the system enables the precise identification of implicit disaster chains and the predictive modeling of risk propagation.

4.1. Subgraph isomorphism for pattern matching

Meteorological disasters adhere to immutable physical laws (e.g., Heavy Rainfall → Waterlogging → Traffic Paralysis). Consequently, verifying that extracted event sequences align with established domain logic is critical.

Formal Definition and Matching Constraints. We formulate this verification process as a Subgraph Isomorphism problem. By mapping the current Dynamic Event Graph (G_{deg}) onto the static template Disaster Knowledge Graph (G_{dkg}), we filter noise and rectify logical inconsistencies. Formally, let $G_{deg} = (V_{deg}, E_{deg}, L_{deg})$ represent the graph of entities extracted from heterogeneous multi-source textual evidence (e.g., specific heavy precipitation events), where L_{deg} denotes node attributes. Simultaneously, let the static template Disaster Knowledge Graph be defined as $G_{dkg} = (V_{dkg}, E_{dkg}, \Sigma)$, where Σ constitutes the set of physically valid event types. We seek an injective mapping function $\phi : V_{deg} \to V_{dkg}$ such that for any edge $(u, v) \in E_{deg}$, the following constraint holds:

$$(\phi(u), \phi(v)) \in E_{dkg} \quad \text{s.t.} \quad S(L_{deg}(u), \Sigma(\phi(u))) \geq \theta \quad (5)$$

Here, $S(\cdot)$ denotes a semantic similarity metric, and θ represents the matching threshold.

Topology-Aware Dynamic Graph Retrieval. Solving this NP-Hard problem requires efficiency; thus, we employ an enhanced Topology-Aware GraphRAG mechanism. Unlike exact matching algorithms such as VF2, our approach permits approximate matching constrained by Graph Edit Distance (GED). Specifically, we identify potential disaster patterns by computing the topological similarity between local

substructures of G_{deg} and the standard disaster chain templates encoded within G_{dkg}. This approach serves two critical functions: it eliminates causal hallucinations (e.g., the illogical Waterlogging → Rainstorm edge) and structurally completes missing key nodes based on ontological priors, thereby enhancing the reliability of the reasoning anchor.

4.2. Path reasoning for chain discovery

Following pattern matching, the primary objective shifts to inferring potential cascading effects from the complex network structure—specifically, disaster chain discovery. Conventional single-hop reasoning fails to capture long-range cascading risks; to address this, we implement a path reasoning algorithm based on Transitive Closure.

Path Confidence Metric. Let $P = \langle v_1, e_1, v_2, \dots, e_{k-1}, v_k \rangle$ denote a disaster propagation trajectory, where v_i represents an event node and e_i denotes a causal or temporal relation. To quantify the propagation probability along this trajectory, we define the Path Confidence Score $Score(P)$:

$$Score(P) = \prod_{i=1}^{k-1} w(e_i) \cdot \left(\frac{1}{1 + \exp(-\alpha \cdot (T(v_{i+1}) - T(v_i)))} \right) \tag{6}$$

Here, $w(e_i)$ represents the prior causal strength of edge e_i derived from the DKG, $T(\cdot)$ indicates the event timestamp, and α serves as the temporal decay factor. This formulation integrates both logical causality and chronological sequentiality, effectively pruning invalid reverse-temporal paths.

Dual-Stage Reasoning Mechanism. We employ a biphasic inference process:

1. **Explicit Chain Verification:** Subjects causal relations directly cited in the text to logical consistency checking against DKG axioms.
2. **Implicit Risk Deduction:** The algorithm performs a Bounded Depth-First Search (DFS) on unconnected node pairs (v_{start}, v_{end}) to identify candidate trajectories satisfying $Score(P) > \delta$.

This mechanism enables the system to deduce candidate trajectories of secondary disasters that lack explicit textual evidence yet maintain physical plausibility under the established constraints and thresholds. For instance, given the seemingly disjoint entities "Extreme Rainstorm" and "Aging Community", the model can reconstruct the extended causal chain: Rainstorm → Drainage Failure → Building Collapse. In scenarios where evidence is sparse or routing constraints are absent, the system may excessively link temporally adjacent events. Ultimately, the aggregated set of verified paths forms the final disaster evolution logic chain, furnishing the downstream Large Language Model with a reasoning foundation grounded in physical interpretability.

5. Graph-guided context engineering

To bridge the semantic gap and mitigate the logical hallucinations inherent in open-domain LLM generation, we establish a distinct neuro-symbolic interaction layer. Departing from conventional, rudimentary prompt concatenation, we introduce a Graph-Guided Context Engineering mechanism. This mechanism transforms the structured implicit risk subgraphs (G_{sub}) retrieved by upstream modules into strict, executable logical constraints. By integrating topological serialization with ontology constraint injection, we compel the model to transition from stochastic text generation to rigorous logical causal reasoning.

5.1. Path-to-text template generation via topological serialization

The retrieved Dynamic Event Graphs encode the critical topology of disaster evolution; however, naïve conversion into natural language often dilutes essential causal and temporal fidelity. To address this, we introduce a serialization protocol designed to preserve topological dependencies, mapping the graph structure G_{sub} into a linearized, context-rich text representation $T_{context}$.

DFS-based Topological Serialization. Given a retrieved isomorphic subgraph $G_{sub} = (V_{sub}, E_{sub})$, our algorithm transcends simple node traversal by explicitly encoding the causal logic and temporal latency embedded within the edges. Formally, we transform any verified risk propagation trajectory $P = \langle v_1, e_1, v_2, \dots, v_k \rangle$ into a structured descriptor compliant with JSON Schema. The serialization function $\Phi(P)$ is defined as:

$$\Phi(P) = \bigoplus_{i=1}^{k-1} Template(v_i, e_i, v_{i+1}, \Delta t_i) \tag{7}$$

Here, $\bigoplus$ denotes semantic concatenation. Beyond mere factual statements, the generated templates incorporate probabilistic markers derived from Bayesian path confidence.

Template Construction. Consider the trajectory: Heavy Rainfall → River Level Surge → Subgrade Hollowing. Rather than generating unstructured prose, the system outputs structural units enriched with logical metadata, as illustrated in Fig. 2.

By explicitly preserving the logical dependencies between (v_i, v_j), this serialization strategy provides the LLM with a structured reasoning scaffold and mitigates the loss of intermediate causal links during extended text generation. This failure mode is analogous to catastrophic forgetting in continual learning [60,61].

5.2. Context-aware chain generation with constraint injection

Upon acquiring the serialized graph context, the primary objective is to construct a generative environment governed by strict logical constraints. We employ an Ontology-Guided Structured Prompting paradigm. This paradigm reframes the LLM not as an unconstrained creative agent, but as a neuro-symbolic executor operating within strict inferential boundaries. To distinguish this stage from the extraction-phase OGPE detailed in Section 3.1, we denote the corresponding generative structures as G_{role}, G_{schema}, and G_{rule}. While these layers share the identical role prior, ontology slots, and rule inventory, they process serialized subgraph contexts instead of raw text. Specifically:

1. **Cognitive Role Layer** (G_{role})**:** Activates the latent semantic space of the model as a meteorological disaster expert, defining the objective to move beyond description and infer implicit causal chains from the provided graph context.
2. **Schema Instantiation Layer** (G_{schema})**:** The canonical types, required slots, and serialized path contexts specified by P_{Ω} are transformed into generation inputs. This process ensures that the resulting output chain strictly complies with the structural constraints associated with nodes, relations, and fields.
3. **Rule-and-Output Layer** (G_{rule})**:** The predicates C_{temp}, C_{causal}, and $C_{intensity}$ are converted into natural language conditions. Subsequently, these conditions are integrated with JSON output instructions and motif completion requirements, thereby constraining the generation and subsequent screening of candidate chains.

By integrating graph context with PNI constraints, the prompt functions as a "soft program" that constrains the semantic search path via physical laws, thereby reducing logical fallacies at their source.

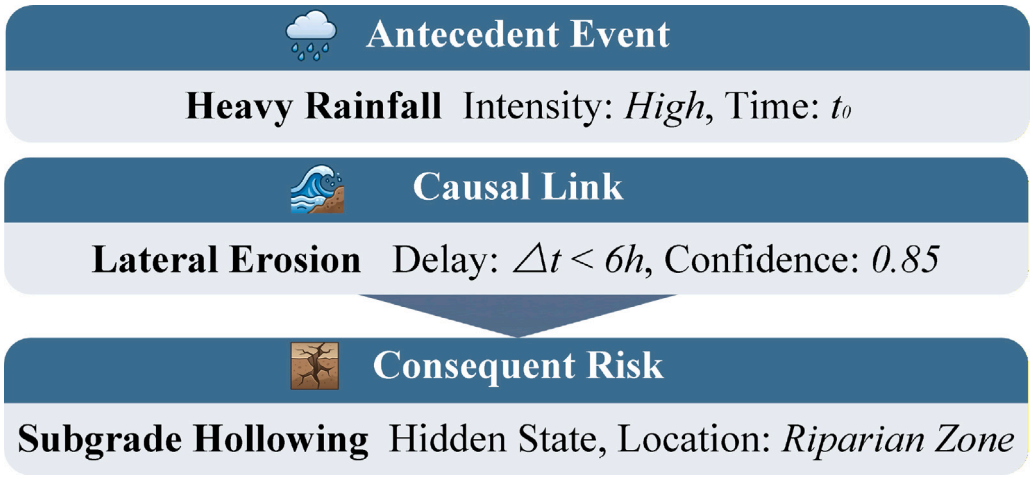


Fig. 2. Topology-serialization blocks. Each block records the event, temporal lag, confidence, and physical state along a candidate chain.

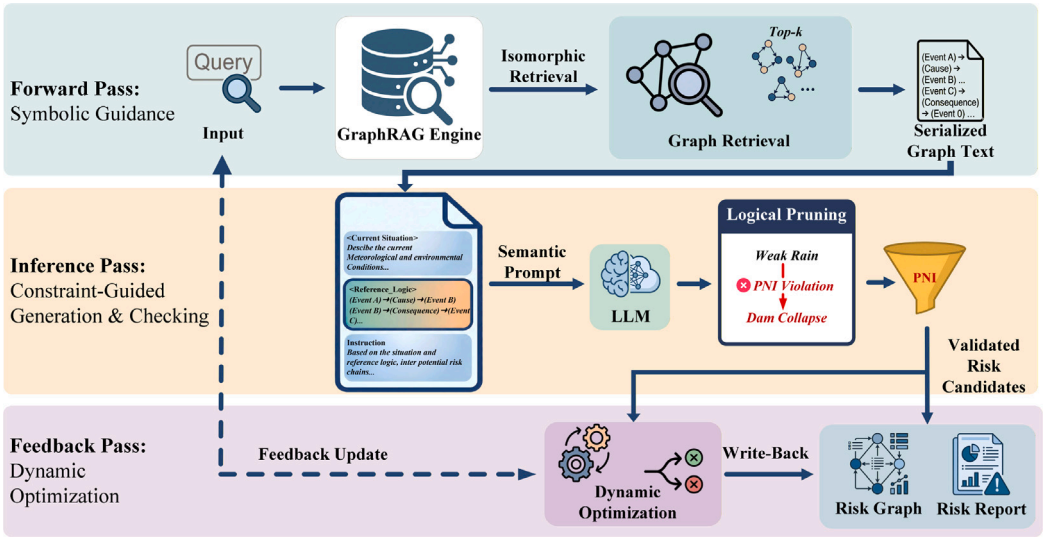


Fig. 3. Neuro-symbolic closed loop. The workflow includes symbolic guidance, constraint-guided generation and checking, and dynamic optimization.

5.3. End-to-end synergy: The Neuro-Symbolic Closed Loop

These modules integrate to form a Neuro-Symbolic Closed Loop, connecting initial data perception to final logical reasoning (Fig. 3).

Forward Pass: Symbolic Guidance. The workflow initiates via the Topology-Aware GraphRAG module, which employs subgraph isomorphism algorithms to retrieve historical cases (G_{sub}) sharing similar evolutionary topologies. Using the serialization protocol, this symbolic knowledge transforms into structured prompts that steer the LLM in generating preliminary risk analysis reports. Here, symbolic knowledge serves as a logical anchor for the neural network, thereby reducing the high-entropy search space.

Inference Pass: Constraint-Guided Generation and Consistency Checking. During the inference phase for constraint-guided generation and consistency checking, G_{rule} serves as the instantiated execution interface of PNI. It incorporates temporal, spatial, and intensity constraints as structured conditions to guide the exploration of the search space for candidate quadruples and chains. Following generation, an external deterministic symbolic validator evaluates candidate edges and complete paths by applying C_{temp}, C_{causal}, and $C_{intensity}$, thereby discarding outputs that violate physical or logical rules. Only validated candidates proceed to the subsequent write-back and analysis stages.

Feedback Pass: Dynamic Optimization. Following structural parsing and consistency checking, the generated risk reports are incorporated into the DCKG as new nodes or edges, thereby updating the current disaster state $h_v(t)$. This stage functions as an event-driven correction process. If a newly integrated report fails the preceding consistency checks or introduces a Bayesian conflict with subsequent evidence, the system initiates a conflict resolution mechanism, rectifies the affected graph connections, and propagates these instances to the retrieval module as negative samples. This iterative process terminates when no new conflicts are detected or when a predefined execution threshold is reached.

To elucidate the operational mechanism of this feedback path, Fig. 4 decomposes the correction routine into numbered steps and decision nodes. Initially, a candidate chain undergoes a check for PNI violations. If no violation is detected, the chain is directly incorporated into the validated risk graph. Conversely, if a violation is detected, the system sequentially executes Bayesian conflict resolution, identifies the

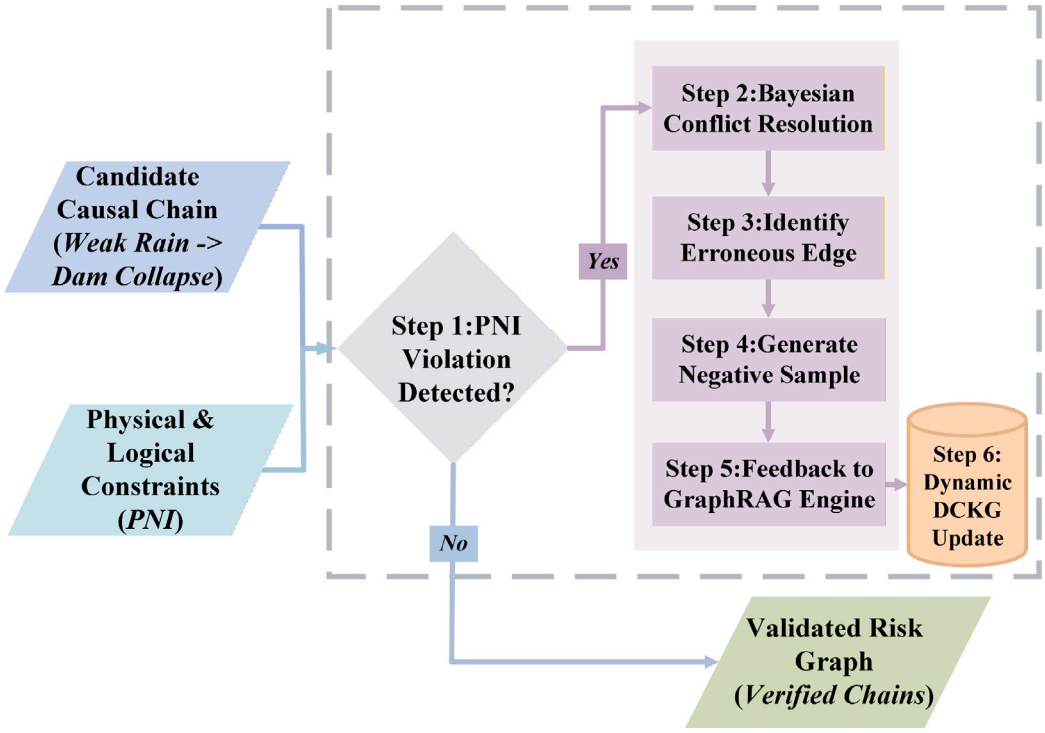


Fig. 4. Closed-loop feedback correction. A candidate chain is either accepted or sent through conflict resolution, error localization, hard-negative generation, GraphRAG feedback, and DCKG update.

erroneous edge, generates a negative retrieval sample from the affected trajectory, provides feedback to the GraphRAG engine, and dynamically updates the DCKG.

6. Experiments

6.1. Experimental setup

6.1.1. Dataset

To validate the efficacy of NeuroSymbolic-Chain in parsing complex meteorological disaster chains, we constructed and evaluated the Multi-source Disaster Semantic Corpus (MDSC). The MDSC dataset comprises 103,088 textual samples, partitioned into 46,747 training samples, 34,127 validation samples, and 22,214 test samples. Owing to the inclusion of restricted operational materials from meteorological agencies and third-party textual records within the raw multi-source corpus, we delineate the boundaries of data access and the scope of the released derived reproducibility artifacts in the Data and Code Availability section. Data sources are highly heterogeneous, selected to mirror real-world complexity:

1. **Social Media Feeds:** Real-time disaster posts characterized by high noise levels and subjective sentiment.
2. **Government Emergency Logs:** Semi-structured official response bulletins.
3. **News Media Reports:** Unstructured texts providing in-depth disaster analysis.

Given the inherent subjectivity in the extraction of complex disaster chains, the construction of the ground truth for the test set underwent a rigorous validation process. Two independent domain experts annotated a random subset of 2000 records. Inter-annotator agreement was measured using Cohen's kappa coefficient (κ). The κ scores for entity boundary extraction, relation typing, and full chain topological matching were 0.88, 0.85, and 0.81, respectively. These values indicate almost perfect agreement, ensuring that the gold standard for the evaluation of all models is highly reliable and consistent. Any remaining discrepancies were resolved through discussion and final adjudication by a third senior expert in meteorology, thereby ensuring the authoritative quality of the benchmark.

To capture the temporal drift inherent in real-world disaster evolution and rigorously preclude data leakage, we rejected conventional random partitioning in favor of a strict time-series split strategy (Table 2). Comprehensive details regarding the redistribution constraints of the raw textual corpus and the release scope of the derived reproducibility artifacts are provided in the Data and Code Availability section.

Statistical analysis reveals a distribution characterized by high-frequency, high-impact events and a significant "long tail" of compound disaster chains. Fig. 5 preserves the original count-based yearly trajectories and the economic-loss envelope, enabling cross-year comparison without obscuring the absolute event scale. The category-level statistical summary is reported separately in Table 3 to improve readability.

6.1.2. Baselines

We benchmark NeuroSymbolic-Chain against eight representative text-based baselines, organized in Table 4 by evidence source, context construction, and governing mechanism. These baselines span closed-book prompting, dense retrieval, temporal retrieval, and graph-based RAG variants. We additionally include the EPA Storm Water Management Model (SWMM) — a widely adopted hydrologic-hydraulic simulator that solves the Saint-Venant equations across urban drainage networks — as a specialized non-generative reference under the shared event-level protocol.

The prompt architecture illustrated in Fig. 6 is proposed specifically for the extraction and generation of meteorological disaster chains. The design follows established paradigms for structured outputs and ontology-guided prompting [14,15]. It also draws on prior work on knowledge-graph prompting and structure-guided prompting [29,36].

To empirically substantiate the structural characteristics of the inferred knowledge base, Fig. 7 presents a macroscopic visualization of the entire KG constructed from the public training set (comprising

Table 2
Chronological dataset split ($N = 103,088$).

Subset	Period	Samples	Share (%)	Primary role
Training	2000–2018	46,747	45.3	Model fine-tuning and ontology knowledge injection
Validation	2019–2022	34,127	33.1	Hyperparameter selection and threshold calibration
Test	2023–2025	22,214	21.5	Strict forward-only held-out evaluation

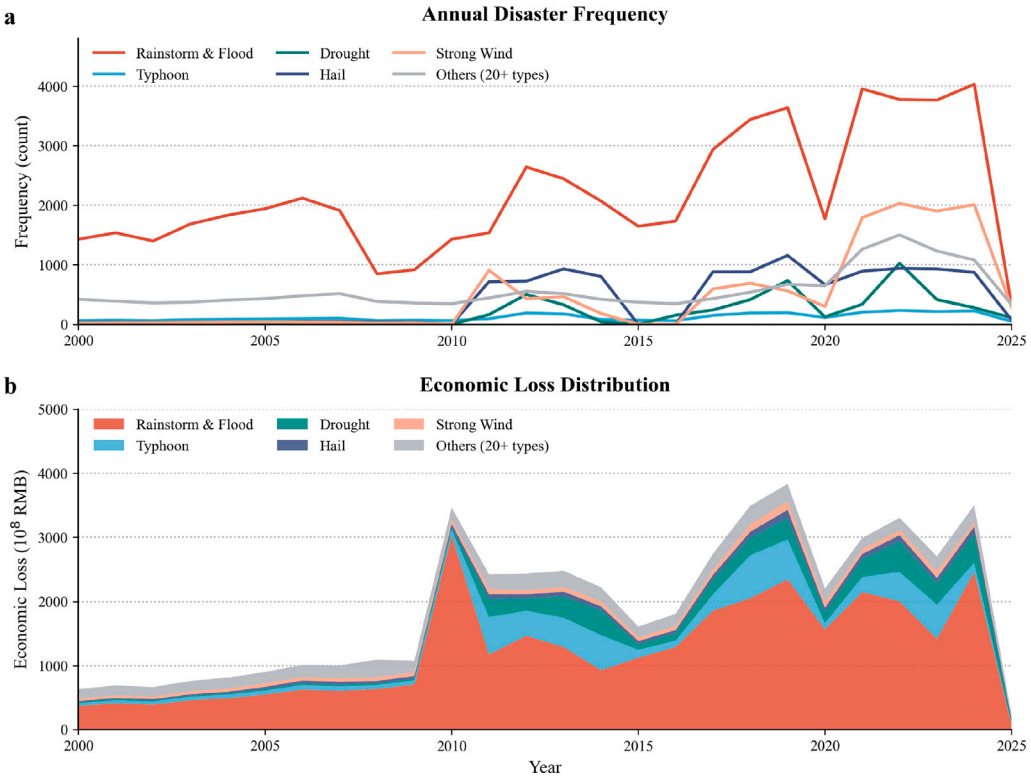


Fig. 5. Yearly disaster counts and economic-loss envelope of the MDSC corpus (2000–2025). The curves preserve the original count-based yearly trajectories, enabling cross-year comparison without obscuring the absolute event scale.

Table 3
Category-level summary of the MDSC corpus.

Category	Total	Avg./yr	Peak year	Loss (10^8 RMB)	Loss share (%)
Rainstorm & Flood	56,751	2183	2024	31,451.7	62.8
Typhoon	3,132	120	2018	5,947.0	11.9
Drought	5,056	194	2022	4,239.1	8.5
Hail	10,847	417	2019	1,452.2	2.9
Strong wind	12,470	480	2022	1,733.6	3.5
Snow	2,269	87	2018	820.3	1.6
Freeze damage	2,251	87	2018	1,006.3	2.0
Chilling injury	1,846	71	2018	729.2	1.5
Geological disaster	2,031	78	2019	1,093.7	2.2
Continuous rain	1,014	39	2017	446.4	0.9
Other	796	31	2018	488.9	1.0
Weather-related rumors	252	10	2012	315.4	0.6
Cold wave	580	22	2013	111.8	0.2
Tornado	162	6	2019	71.3	0.1
Heat wave	266	10	2017	70.9	0.1
Thunderstorm	1,912	74	2011	28.7	0.1
Weather Dept. emergency Res.	53	2	2012	32.8	0.1
Forest & Grassland fire	788	30	2019	12.5	0.0
Heavy fog	208	8	2011	9.3	0.0
Sandstorm	309	12	2023	10.1	0.0
Ice jam	19	1	2012	8.3	0.0
Storm surge	25	1	2011	1.7	0.0
Hazardous material leakage	15	1	2000	0.0	0.0
Dry hot wind	20	1	2014	0.0	0.0
Air pollution	16	1	2014	0.0	0.0
Total	**103,088**	**3965**	–	**50,081.2**	**100.0**

Note: Loss share is calculated based on the total direct economic loss.

Table 4
Baseline taxonomy.

System	Paradigm	Evidence source	Context construction	Governing mechanism
Text-based baselines and proposed framework				
Pure LLM	Closed-book	Parametric memory	Direct prompting	No external evidence or constraints
Vanilla RAG	Dense RAG	Dense Top-5 passages	Flat prompt concatenation	Retrieval-only baseline
ChronoRAG	Temporal RAG	Time-weighted passages	Flat prompt concatenation	Temporal decay reweighting
GraphRAG	Graph RAG	Community summaries	Graph-derived summaries	Community-structured retrieval
RAPTOR	Hierarchical RAG	Recursive tree nodes	Hierarchical abstraction	Tree-based recursive retrieval
HippoRAG	Memory RAG	KG-guided recall	Associative memory context	Multi-hop memory traversal
LightRAG	Dual-level graph RAG	Graph index + dual retrieval	Dual-level graph context	Dual-granularity retrieval
DyG-RAG	Dynamic graph RAG	Dynamic relation graph	Time-updated graph context	Relation-evolution refresh
OGPE-only	Diagnostic variant	Source text only	Ontology-guided prompt	No graph reasoning
Ours	Neuro-symbolic	Dynamic subgraph retrieval	Topology serialization	PNI physical constraints
Specialized non-generative reference				
SWMM	Process simulator	Rainfall, terrain, and drainage inputs	Hydraulic state evolution	Saint-Venant equations

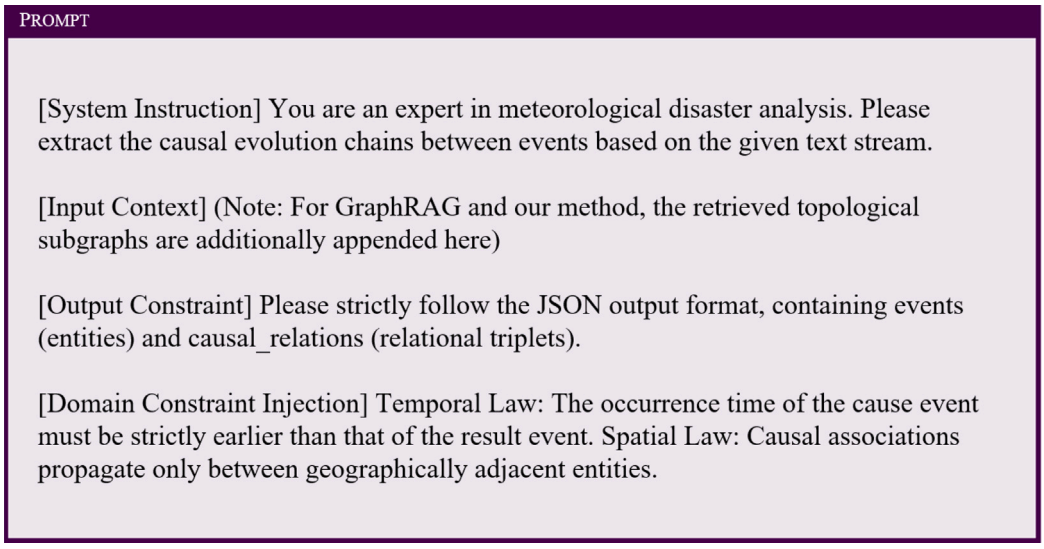


Fig. 6. Constraint-guided prompting. Task slots, context slots, and output schema are paired with physical rules.

99,026 de-identified nodes and 127,940 directed triples), alongside a magnified local disaster chain representing a typical compounding hazard sequence.

To complement this visual representation, Fig. 8 delineates the corresponding quantitative distributions, including the ontological composition of nodes, the inventory of relations, and the provenance of evidence. Together with the component statistics reported above, these visual and quantitative summaries substantiate the scale and structure of the constructed training KG.

6.1.3. Implementation details

Qwen3-32B [62] is adopted as the default backbone for the primary controlled experiments because it showed the most stable performance in preliminary screening for Chinese meteorological entity parsing and JSON-constrained generation. The cross-model comparison is reported in Table 9. Hyperparameters are fixed to a 4096-token budget, temperature 0.5, and Top-p 0.9. For the text-based baselines, the backbone, token budget, and evaluation pipeline are kept fixed. The graph-based baselines retain their native retrieval logic and context organization and are normalized only at the final JSON output stage. The feedback loop is capped at three rounds, and most triggered cases terminate within two rounds. The final configuration file, core ontology definitions, prompt templates, split metadata, and evaluation utilities are fully documented in Appendices A.1–A.3 and the Data and Code Availability section, and are hosted within our public reproducibility repositories.

6.1.4. Latency evaluation protocol

To avoid ambiguity in efficiency evaluation, we clarify the latency values reported in this study according to four measurement contexts. First, isolated per-sample end-to-end latency measures the complete inference time for a single sample, including retrieval, graph reasoning, LLM decoding, and method-native corrective iterations. This metric is used for direct comparison among text-based systems under the same hardware, backbone, token budget, and runtime protocol. Second, graph-stage latency isolates graph I/O, bounded DFS, subgraph matching, deterministic symbolic validation, and Bayesian feedback, and is used only to localize the computational overhead of NeuroSymbolic-Chain rather than to replace the full E2E latency. Third, streaming-load latency additionally includes queueing and concurrent request handling under bursty input streams, and therefore reflects operational deployment conditions rather than isolated model inference. Finally, the runtime of SWMM is reported only as cross-paradigm operational context, because SWMM performs grid-level hydraulic simulation over structured physical inputs, whereas NeuroSymbolic-Chain performs text-driven inference over heterogeneous textual evidence. Accordingly, only isolated per-sample E2E latency is treated as the direct efficiency comparison among text-based systems.

6.1.5. Evaluation metrics

To rigorously benchmark the performance of NeuroSymbolic-Chain in parsing complex meteorological disaster chains, we organize the evaluation into five complementary levels: local extraction fidelity (Entity F1 and Relation F1), strict end-to-end chain correctness (Chain

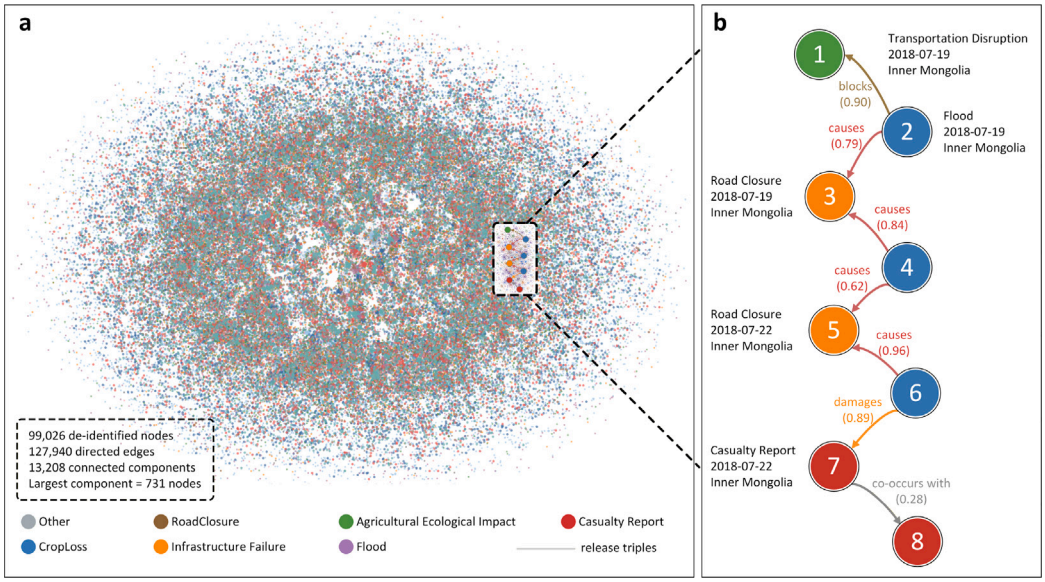


Fig. 7. Training KG layout. (a) Global view with a representative local chain boxed; (b) magnified view of the boxed chain.

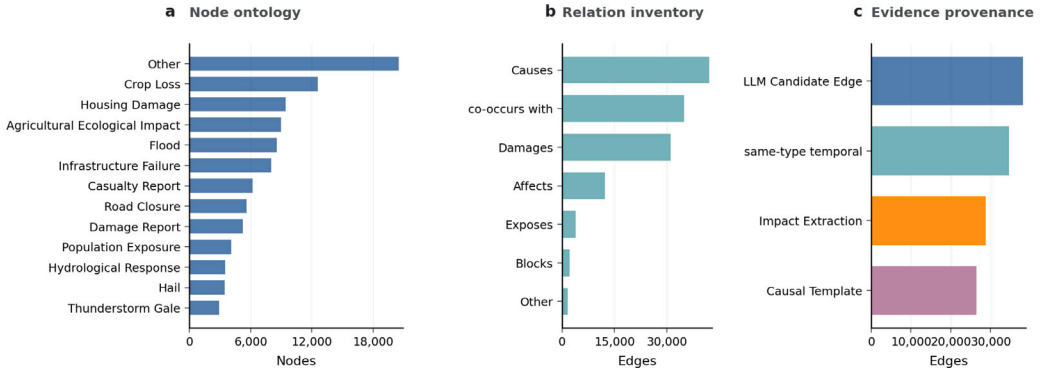


Fig. 8. KG statistics: (a) node ontology composition, (b) relation inventory, and (c) evidence provenance.

Accuracy), rule-governed logical validity (LVR), evidence-grounded factual consistency (Fact Consistency), and a shared event-level benchmark protocol. Disaster chain analysis involves correctness at the node, relation, path, and report levels.

1. Extraction Performance (Entity F1 & Relation F1): We gauge the precision of entity recognition and relation extraction via standard F1 scores. Given a prediction set P and ground truth set G, we define Precision (P), Recall (R), and F1 scores as:

$$P = \frac{|P \cap G|}{|P|} \tag{8}$$

$$R = \frac{|P \cap G|}{|G|} \tag{9}$$

$$F1 = \frac{2 \cdot P \cdot R}{P + R} \tag{10}$$

Here, Entity F1 quantifies the accurate identification of disaster nodes (e.g., Rainstorm, Traffic Paralysis). Conversely, Relation F1 evaluates triplet (h, r, t) accuracy, counting a prediction as correct solely if the head entity, tail entity, and relation type simultaneously match the ground truth.

2. Chain Accuracy: To assess the capacity of the model for long-chain reasoning, we introduce Chain Accuracy. Given a ground truth disaster chain $C_{gold} = \langle e_1 \xrightarrow{r_1} e_2 \ldots \xrightarrow{r_{k-1}} e_k \rangle$ of length k, a prediction C_{pred} is deemed correct only if it achieves exact topological and semantic alignment with C_{gold}:

$$Acc_{chain} = \frac{1}{N} \sum_{i=1}^{N} I(C_{pred}^{(i)} \equiv C_{gold}^{(i)}) \tag{11}$$

where N is the total sample size and $I(\cdot)$ is the indicator function. This metric is stricter than isolated relation extraction because any break or error in an intermediate causal link is penalized. However, a single valid disaster chain can be represented at varying granularities in natural language (e.g., Rainfall → Flooding versus Rainfall → River Surge → Flooding), and the explicit annotation of intermediate nodes often depends on established annotation boundary conventions.

To reward semantically valid but topologically non-identical predictions, we additionally compute three complementary metrics: Path Similarity (denoted as PathSim and reported as P-Sim in result tables), Normalized Edit Distance (NED), and Relative Root Mean Square Error (RRMSE).

$$PathSim(C_{pred}, C_{gold}) = \frac{2|A(C_{pred}, C_{gold})|}{|C_{pred}| + |C_{gold}|} \tag{12}$$

$$NED(C_{pred}, C_{gold}) = \frac{ED(C_{pred}, C_{gold})}{\max(|C_{pred}|, |C_{gold}|)} \tag{13}$$

$$RRMSE = \sqrt{\frac{1}{N} \sum_{i=1}^{N} \left(\frac{l_{i,pred} - l_{i,gold}}{\max(l_{i,gold}, 1)} \right)^2} \tag{14}$$

Here, $A(C_{pred}, C_{gold})$ denotes the set of ontology-aligned nodes and relations shared by the predicted and gold chains, $ED(\cdot,\cdot)$ denotes the minimum edit cost between two chains, and l_i is the chain length of sample i. Path Similarity captures partial topological and semantic overlap, NED measures structural editing cost, and RRMSE reports normalized chain-scale deviation. These complementary metrics do not eliminate the inherent subjectivity of the gold annotations. Instead, they mitigate the over-penalization of exact-match evaluation when a prediction preserves the core causal topology but differs from the annotated chain in granularity or the explicit annotation of intermediate nodes. Specifically, Path Similarity and NED serve as the primary structural signals, while RRMSE provides an auxiliary signal for chain-scale deviation.

3. Logic Violation Rate (LVR): To quantify the physical and logical self-consistency of generated content, we propose the Logic Violation Rate (LVR). This metric tracks the frequency of violations against the domain constraint set $C = \{C_{temp}, C_{causal}, C_{intensity}\}$:

$$LVR = \frac{1}{|Y|}\sum_{y\in Y} I(\exists c \in C, Violates(y,c)) \times 100\% \tag{15}$$

Here, Y denotes the set of all generated causal assertions. The boolean function $Violates(y,c)$ returns 1 if assertion y breaches constraint c (e.g., violating the temporal axiom $t_{cause} > t_{effect}$ or energy conservation principles). Lower LVR values indicate superior physical interpretability and logical robustness.

4. Fact Consistency (FC; reported as F-Con in result tables): To detect generative hallucinations, we deploy GPT-5 as an "LLM-as-a-Judge" [63] to perform evidence-based consistency scoring. Given a generated report R and retrieved source documents D_{src}, the FC score is formulated as:

$$FC(R, D_{src}) = \mathbb{E}_{q\in Q(R)}[Verify(q, D_{src})] \tag{16}$$

where $Q(R)$ represents the set of atomic fact assertions extracted from R. We first require the judge to classify each atomic assertion q into one of three categories: Supported, Unsupported, or Contradicted. Subsequently, we binarize this assessment by setting $Verify(q, D_{src}) = 1$ strictly for Supported claims, and 0 otherwise. We use Chain-of-Thought (CoT) prompting [64] to improve adjudication.

5. Shared Event-Level Evaluation Metrics: To enable a fair comparison between the text-chain parsing framework and physical-process simulation systems, we conduct the evaluation within a unified label space of key urban risk events. Due to the fundamental differences in input modalities and intermediate representations between these paradigms, we restrict the comparative analysis to the outcomes of downstream events. To ensure the transparency of the evaluation, we formally define these metrics. Let E_{pred} and E_{true} denote the sets of predicted risk events and ground-truth risk events, respectively. The risk-hit rate, event recall, and false-alarm rate systematically quantify the precision of alerts, the coverage of actual occurrences, and the proportion of incorrect warnings. These metrics are formulated as follows:

$$\text{Risk-hit Rate} = \frac{|E_{pred} \cap E_{true}|}{|E_{pred}|} \tag{17}$$

$$\text{Event Recall} = \frac{|E_{pred} \cap E_{true}|}{|E_{true}|} \tag{18}$$

$$\text{False-alarm Rate} = \frac{|E_{pred} \setminus E_{true}|}{|E_{pred}|} \tag{19}$$

Furthermore, warning lead time evaluates alert timeliness by averaging the temporal difference between the actual occurrence of an event and the generation of the corresponding warning. To ensure comparability across paradigms, time zero (T_0) is defined as follows: for SWMM, T_0 denotes the moment the input rainfall sequence (forecast or nowcast) is delivered to the simulator; for NeuroSymbolic-Chain, T_0 denotes the moment the system receives the first valid precursor text signal (e.g., a social media post reporting localized waterlogging). The endpoint for both systems is the actual occurrence time of the target event. Finally, average response latency measures computational efficiency, defined as the average wall-clock time required to process input data and generate final predictions. Notably, the response times of the two systems reflect fundamentally different computational paradigms: the latency of NeuroSymbolic-Chain covers only text processing and graph inference overhead, whereas the latency of SWMM stems from grid-level hydraulic integration governed by the Saint-Venant equations. Consequently, these metrics do not constitute a direct performance comparison.

To quantify efficiency within the text-based paradigm, we report the average isolated end-to-end latency, peak GPU memory, total tokens per sample, and total FLOPs per sample under a unified runtime protocol. Because these text-based overhead metrics are evaluated per sample under identical hardware, backbone models, token-budget policies, and execution settings, they provide a direct computational comparison between NeuroSymbolic-Chain and the other text-based baselines. To provide a detailed analysis of the additional computational overhead, we distinguish the isolated per-sample E2E latency from the graph-stage latency. Specifically, within the Main Results subsection, we employ a graph-scale audit to isolate the computational costs associated with approximate matching, path search, and end-to-end graph reasoning. The complete evaluation scripts and programmatic audit utilities are released within the public code repository. The statistical testing protocol for the primary metrics is described in Section 6.1.6.

6.1.6. Statistical significance and uncertainty estimation

To provide a more rigorous assessment of statistical robustness, we conducted paired significance tests and uncertainty estimation for the main comparisons. Because all competing models were evaluated on the same held-out test split, paired tests were used throughout the statistical analysis.

For aggregate continuous or score-based metrics, including Entity F1, Relation F1, Path Similarity, Normalized Edit Distance, Relative Root Mean Square Error, and Fact Consistency, we used paired non-parametric bootstrap tests with 10,000 resampling iterations. In each bootstrap iteration, test samples were resampled with replacement, and the paired metric difference between NeuroSymbolic-Chain and the strongest competing baseline was recomputed. The empirical distribution of the paired differences was then used to estimate 95% confidence intervals and two-sided p-values.

For sample-level binary metrics, including Chain Accuracy and the binary logic-violation indicator used to compute Logic Violation Rate, we used McNemar's tests based on paired disagreement counts between NeuroSymbolic-Chain and the strongest competing baseline. Confidence intervals for these binary metrics were estimated using paired bootstrap resampling over the same held-out test instances. For lower-is-better metrics, including NED, RRMSE, and LVR, the improvement was computed as the value of the strongest competing baseline minus the value of NeuroSymbolic-Chain, so that positive differences consistently indicate better performance of the proposed framework.

To account for multiple comparisons within the family of primary metrics, Holm–Bonferroni correction was applied to control the family-wise error rate. The corrected p-values and 95% confidence intervals for the main comparisons are reported in Supplementary Table S11.

6.2. Main benchmark performance

Benchmark Testing. Table 5, Figs. 9, and 10 present a benchmark comparison of disaster-information parsing under different paradigms. The upper block of Table 5 reports the text-based systems and NeuroSymbolic-Chain, while the lower block reports the event-level comparison with SWMM under the shared event-level protocol. The runtime values in the lower block are included only to contextualize the operational roles of the two systems and should not be interpreted as a direct latency benchmark. The results show that NeuroSymbolic-Chain

Table 5
Main benchmark results ($N = 22,214$).

Text-based parsing systems								
Model	Extraction		Chain structure				Reliability	
	E-F1	R-F1	C-Acc	P-Sim	NED↓	RRMSE↓	LVR (%)↓	F-Con
Pure LLM	0.785	0.724	0.682	0.701	0.314	0.271	14.82	0.738
Vanilla RAG	0.762	0.735	0.645	0.683	0.337	0.294	11.45	0.792
ChronoRAG	0.808	0.759	0.678	0.742	0.268	0.229	6.54	0.841
GraphRAG	0.835	0.778	0.724	0.846	0.183	0.157	7.95	0.876
RAPTOR	0.829	0.784	0.730	0.844	0.185	0.160	6.21	0.880
HippoRAG	0.840	0.795	0.736	0.858	0.174	0.148	5.36	0.891
LightRAG	0.846	0.791	0.735	0.872	0.166	0.140	4.96	0.888
DyG-RAG	0.843	0.793	0.741	0.867	0.169	0.143	4.52	0.887
OGPE-only	0.819	0.781	0.702	0.781	0.224	0.191	4.88	0.858
Ours	**0.878****	**0.842****	**0.785****	**0.908****	**0.121****	**0.114****	**2.15****	**0.915****
Gain vs. strongest text baseline	+3.8%	+5.9%	+5.9%	+4.1%	−27.1%	−18.6%	−52.4%	+2.7%

Shared event-level protocol						
System	Primary input	Hit (%)	Recall (%)	FA (%)	Lead (h)[a]	Runtime (s)[b]
SWMM	Structured rainfall, terrain, and drainage networks	62.4	41.2	**4.5**	2.0	~1250
Ours	Multi-source textual evidence + DCKG	**89.2**	**86.5**	6.8	3.5	1.24

** Indicates Holm–Bonferroni-corrected $p < 0.001$ for the comparison between NeuroSymbolic-Chain and the strongest competing text-based baseline for the corresponding metric. Paired non-parametric bootstrap tests were used for aggregate continuous or score-based metrics, including E-F1, R-F1, P-Sim, NED, RRMSE, and F-Con. McNemar's tests were used for sample-level binary metrics, including C-Acc and the binary logic-violation indicator used to compute LVR. Corrected p-values and 95% confidence intervals for the main comparisons are reported in Supplementary Table S11.

[a] Warning lead time: for SWMM, T_0 is the rainfall-sequence delivery time; for Ours, T_0 is the receipt time of the first valid precursor text signal.

[b] Runtime values are reported only as cross-paradigm operational context, not as direct algorithmic latency comparisons. SWMM performs grid-level hydraulic simulation over structured physical inputs, whereas NeuroSymbolic-Chain performs text-driven graph reasoning over heterogeneous textual evidence. Direct latency comparison among text-based systems is reported in Table 6.

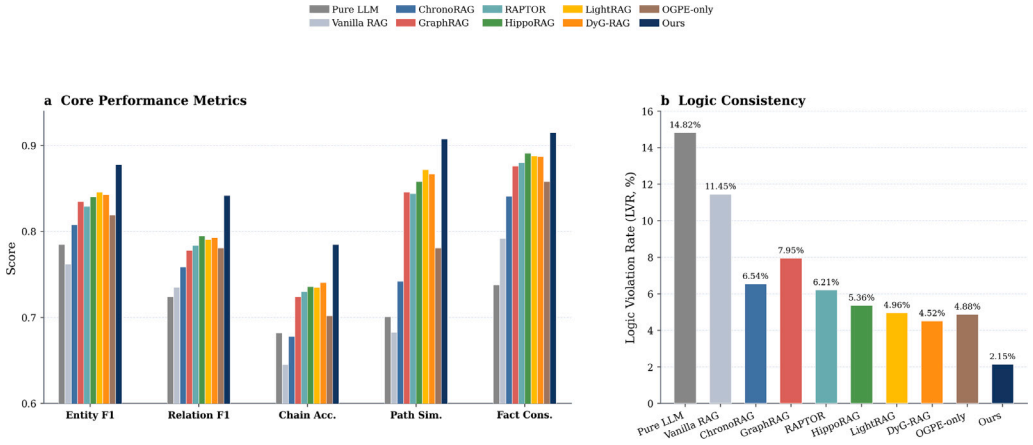


Fig. 9. Core performance and logic consistency across baselines. Panel (a) reports Entity F1, Relation F1, Chain Accuracy, Path Similarity, and Fact Consistency. Panel (b) reports the Logic Violation Rate (LVR) across all text-based systems.

outperforms existing text-chain baselines in strict extraction, partial chain matching, logical consistency, and factual consistency.

The runtime values reported in the shared event-level block of Table 5 serve exclusively to contextualize the respective operational roles of SWMM and NeuroSymbolic-Chain. Consequently, the 1.24-second execution time associated with NeuroSymbolic-Chain represents the latency of the graph stage under this shared event-level protocol, whereas a direct comparison of the isolated per-sample E2E latency among text-based systems is detailed in Table 6.

The accuracy results presented in the upper block of Table 5 can be analyzed from three perspectives. The core performance and LVR results are visualized in Fig. 9, while the normalized eight-metric matrix and relative gains are summarized in Fig. 10.

First, the LVR decreases from 4.52% for DyG-RAG to 2.15%, a relative reduction of 52.4%. Pure LLM reaches 14.82%, indicating that temporal inversion and causal loops are more likely in the absence of constraints. These results demonstrate that PNI and explicit physical constraints effectively enhance logical consistency. Concurrently, under the default Qwen3-32B setting, the Fact Consistency metric reaches 0.915, indicating that the imposition of stronger constraints does not compromise evidential consistency.

Second, after the introduction of complementary metrics that accommodate differences in granularity, the chain-level performance of the model remains robust: Chain Accuracy increases from 0.741 to 0.785, Path Similarity from 0.872 to 0.908, and NED decreases from 0.166 to 0.121. This indicates that the system is insensitive to the subjective boundary variations introduced by manual annotation.

Third, RRMSE decreases from 0.140 to 0.114, indicating that the model more stably preserves the overall path length and causal progression structure. Concurrently, the Entity F1 score increases to 0.878 (+3.8%), and the Relation F1 score reaches 0.842 (+5.9%). These results demonstrate that the framework enhances the stability of both disaster node recognition and relation typing. Specifically, OGPE plays

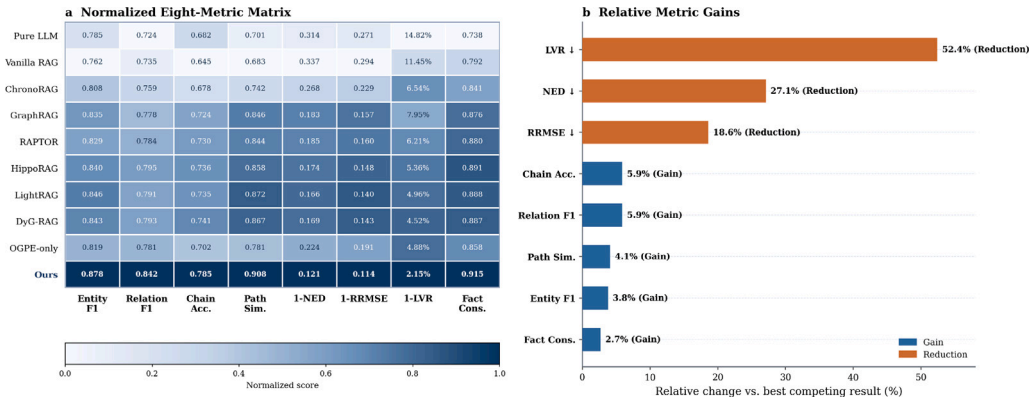


Fig. 10. Normalized metric matrix and relative gains over the strongest competing baseline. Panel (a) shows the normalized eight-metric matrix, where higher normalized values indicate better performance. Panel (b) reports the relative gains or reductions achieved by NeuroSymbolic-Chain.

a critical role in this improvement by resolving referential ambiguity within domain terminology.

6.3. Computational efficiency and trade-off analysis

Trade-off between efficiency and consistency. Table 6 presents a direct comparison of the isolated end-to-end efficiency of the NeuroSymbolic-Chain framework against other text-based systems under a unified hardware environment (1 × NVIDIA A100 80 GB). Compared with DyG-RAG, the NeuroSymbolic-Chain framework incurs an additional per-sample overhead of 0.28 s in end-to-end latency, 1.4 GB in peak GPU memory, 470 tokens, and 30.1 T FLOPs. Under the strict per-sample E2E protocol, the total latency reaches 1.96 ± 0.35 s.

To delineate the sources of this additional computational overhead without conflating the latency of the graph stage with the end-to-end (E2E) latency, we employ a representative graph-scale audit as a baseline for the graph stage. At this operational scale, the graph-reasoning stage requires 1.24 ± 0.38 s, where approximate matching and path search account for 0.362 ± 0.115 s and 0.338 ± 0.110 s, respectively. The remaining execution time within the graph stage is attributed to embedding-cache lookups, graph I/O, deterministic validation, Bayesian feedback, and other orchestration overheads; consequently, we do not further decompose this specific duration into isolated subcomponents. Furthermore, relative to the isolated E2E latency of 1.96 ± 0.35 s reported in Table 6, the time difference between the full E2E process and the graph stage is primarily attributable to LLM decoding and the execution overhead native to the framework.

Furthermore, although Table 6 reports the isolated inference time for a single sample, a stream-load stress test reveals that the latency under the 1× nominal stream-load setting increases to 2.3 s due to additional queueing and concurrent request handling. Under the extreme 8× burst setting, the E2E latency surges to 54.2 s as a result of severe queue buildup and load shedding, while the triggered feedback ratio declines to 6.3%. This observation indicates that complex feedback loops are preempted during overload conditions rather than acting as the primary computational bottleneck. Consequently, this elevated metric reflects a performance limitation under stress and should not be interpreted as the isolated per-sample latency suitable for direct comparison across text-based systems. Collectively, these results elucidate the underlying trade-off: an additional 0.28 s in the isolated end-to-end latency yields a substantial reduction in the LVR from 4.52% to 2.15%, accompanied by an improvement in Chain Accuracy from 0.741 to 0.785.

Consequently, Table 6 provides a direct comparison of the isolated per-sample E2E efficiency among text-based systems, whereas the graph-stage audit serves exclusively to isolate the internal computational overhead associated with NeuroSymbolic-Chain.

6.4. Fine-grained performance across disaster categories

Category-level heterogeneity across major disaster types. To determine whether the aggregate scores in Table 5 obscure domain-specific behavior, Table 7 reports the two chain-level indicators that most directly address the concerns of the reviewer — Chain Accuracy and Logic Violation Rate — for NeuroSymbolic-Chain and DyG-RAG across four primary disaster categories within a profiled evaluation subset derived from the held-out MDSC test split: Rainstorm-Flooding, Strong Wind, Hail, and Drought. These four categories collectively account for 84.83% of this profiled subset. Compared with DyG-RAG, NeuroSymbolic-Chain improves Chain Accuracy by 0.046, 0.042, 0.040, and 0.073 on Rainstorm-Flooding, Strong Wind, Hail, and Drought, respectively, while reducing the LVR by 2.40, 2.05, 2.10, and 4.30 percentage points. The largest gain occurs in the slow-onset Drought category, where Chain Accuracy increases from 0.648 to 0.721 and the LVR decreases from 7.15% to 2.85%. By contrast, the margins are narrower for sudden-onset hazards, such as Strong Wind and Hail, where DyG-RAG remains competitive because precursor signals are denser and causal spans are shorter. This contrast suggests that the primary advantage of NeuroSymbolic-Chain lies in maintaining logically constrained long-latency links when explicit textual anchors are sparse, rather than in saturating short and dense causal chains. Consequently, the category-level analysis indicates that the performance gains are not confined to a single hazard regime, while identifying slow-onset drought as the clearest stress case for future refinement.

We note that category-specific latency is affected by both chain length and evidence density. Therefore, the category-level results are used primarily to diagnose chain accuracy and logical validity across disaster regimes, whereas the unified direct latency comparison among text-based systems is reported in Table 6.

6.5. Operational usability and expert evaluation

Independent Expert Blind Review. A blinded expert audit was conducted on 128 stratified held-out candidate trajectories. Experts labeled 89.8% of the trajectories as physically and logically plausible, corresponding to a residual-risk rate of 10.2% on the audited subset. The DFS-involved subset showed a false-positive rate of 26.2%, substantially higher than the 2.3% observed for explicit-only trajectories. A representative residual failure manifested as a hydrological-lag violation, wherein the system erroneously linked upstream heavy rainfall to a downstream flash flood within an implausibly short propagation window. This boundary condition is consistent with recent evidence that automated judge models remain unstable in factual evaluation [65, 66] and that AI weather systems require explicit checks for physical consistency prior to operational deployment [67,68].

Table 6
Direct isolated end-to-end efficiency comparison among text-based systems.

Model[a]	Latency (s)↓	Mem. (GB)↓	Tokens↓	FLOPs (T)↓	LVR (%)↓
Pure LLM	0.72 ± 0.12	69.2 ± 0.4	450 ± 45	28.8 ± 2.9	14.82
Vanilla RAG	0.95 ± 0.15	70.1 ± 0.5	1250 ± 110	80.0 ± 7.0	11.45
ChronoRAG	0.98 ± 0.16	70.2 ± 0.5	1300 ± 125	83.2 ± 8.0	6.54
GraphRAG	1.35 ± 0.21	71.5 ± 0.8	2450 ± 210	156.8 ± 13.4	7.95
RAPTOR	1.42 ± 0.22	71.8 ± 0.9	2600 ± 230	166.4 ± 14.7	6.21
HippoRAG	1.51 ± 0.25	72.1 ± 0.7	2750 ± 190	176.0 ± 12.2	5.36
LightRAG	1.55 ± 0.24	72.3 ± 0.8	2820 ± 185	180.5 ± 11.8	4.96
DyG-RAG	1.68 ± 0.28	72.8 ± 0.9	2950 ± 205	188.8 ± 13.1	4.52
OGPE-only	0.88 ± 0.14	69.6 ± 0.5	850 ± 75	54.4 ± 4.8	4.88
Ours	**1.96 ± 0.35**[b]	**74.2 ± 1.2**	**3420 ± 415**	**218.9 ± 26.5**	**2.15**

[a] All values are measured as isolated per-sample end-to-end runtime statistics on a single NVIDIA A100 80 GB Tensor Core GPU over $N = 22,214$ test samples.

[b] The reported E2E latency includes graph-stage routing, LLM decoding, and method-native corrective iterations. FLOPs are approximate decoder-side estimates calculated as $2 \times 32B\times$ total tokens using the fixed Qwen3-32B backbone.

Table 7
Fine-grained chain evaluation on the MDSC test subset.

Category	DyG-RAG		Ours	
	C-Acc	LVR (%)↓	C-Acc	LVR (%)↓
Rainstorm-flooding	0.742	4.60	**0.788**	**2.20**
Strong wind	0.770	3.80	**0.812**	**1.75**
Hail	0.765	3.95	**0.805**	**1.85**
Drought	0.648	7.15	**0.721**	**2.85**

Note: Table 7 focuses on the two chain-level indicators that most directly address the concerns of the reviewer, namely Chain Accuracy and Logic Violation Rate. DyG-RAG serves as the reference baseline because it represents the strongest overall text baseline regarding the chain-level criteria presented in Table 6. The four categories analyzed in this table — Rainstorm-Flooding (53.21%), Strong Wind (13.58%), Hail (13.31%), and Drought (4.73%) — collectively account for 84.83% of the profiled evaluation subset derived from the held-out test split.

Table 8
Expert utility ratings.

Task scenario	Mean rating	Representative operational value
Early risk prompting	4.1 ± 0.7	8/10 experts rated latent-risk prompting useful before explicit failures appeared
Event-level situational analysis	4.3 ± 0.6	Supports event-level assessment under fragmented evidence
Mechanism explanation	4.6 ± 0.5	Provides causally coherent explanations aligned with physical evolution

Specialized Non-Generative System Comparison. SWMM is driven by structured inputs of rainfall, terrain, and drainage networks, whereas NeuroSymbolic-Chain reasons over fragmented, multi-source textual evidence; consequently, the time zero (T_0) of the latter inherently differs from that of a rainfall-driven simulator (see Section 6.1). Therefore, both systems are mapped to a shared downstream label space centered on key urban risk events. Under this shared evaluation protocol, NeuroSymbolic-Chain, functioning as a text-driven proxy system, exhibits higher event-level recall (86.5% versus 41.2%) and a higher risk-hit rate (89.2% versus 62.4%) for events with observable textual precursors. However, SWMM demonstrates a lower false-alarm rate (4.5% versus 6.8%), confirming the irreplaceable advantage of enforcing strict physical laws to suppress spurious alerts. The response latency of NeuroSymbolic-Chain (1.24 s) reflects the lightweight overhead of text inference, whereas the substantially higher runtime of SWMM (~1250.0 s) corresponds to grid-level hydraulic integration based on the Saint-Venant equations—metrics that are not directly comparable. These complementary strengths indicate that the two paradigms serve distinct but synergistic roles within an integrated early-warning pipeline, rather than one superseding the other.

Task-Oriented Expert Usability Assessment. To evaluate the practical utility of the framework, the ten-member expert panel assessed the system outputs on the same 128 audited candidate trajectories from this stratified held-out set across three core application scenarios using a fixed 5-point Likert scale (1 = not useful, 5 = highly useful). The evaluation was grounded in real-world meteorological events drawn from the held-out corpus and spanning multiple hazard types, including rainstorms, typhoons, and extreme cold waves. The July 2023 extreme rainstorm in Beijing is retained only as a representative anchor case, and the results are summarized in Table 8. NeuroSymbolic-Chain obtained mean scores of 4.1 ± 0.7 for early risk prompting, 4.3 ± 0.6 for event-level situational analysis, and 4.6 ± 0.5 for mechanism explanation. These results indicate the capacity of the framework to process the fragmented information typical of early disaster response. Furthermore, the panel observed that even when early evidence lacked explicit descriptions of downstream failures, the system successfully inferred structural risks, such as riparian subgrade hollowing, from preliminary precursor signals such as localized waterlogging. Eight of the ten experts rated this predictive capability as useful (≥4). By incorporating the Principles of Narrative Invariance, the framework avoids the logical and temporal errors common in conventional retrieval models and generates mechanistic explanations consistent with the observed physical evolution of the events.

6.6. Cross-model generalization

Cross-Model Generalization and Setup. To investigate whether the observed improvements depend on a specific foundation model, Table 9 presents two complementary blocks following the format of Table 5. The upper block reports the absolute performance of NeuroSymbolic-Chain across sixteen models that span various parameter scales and architectures. By keeping the ontology injection, prompt templates, token budgets, and evaluation pipeline constant while replacing only the final generation module, the results reveal a clear scaling trend within model families: larger or more advanced models yield incremental gains in Chain Accuracy and Fact Consistency. To isolate the contribution of the neuro-symbolic constraint layer from the impact of the backbone model scale, the lower block compares NeuroSymbolic-Chain with the robust GraphRAG baseline, which is selected from Table 5 as a representative structured-retrieval comparator. This evaluation utilizes three distinct backbones — Qwen3-8B, Qwen3-32B, and GPT-4o — under identical token budgets, prompt templates, and evaluation pipelines.

In the controlled same-backbone comparisons against GraphRAG (Table 9, lower block), NeuroSymbolic-Chain consistently demonstrates superior logical consistency. Specifically, it enhances Chain Accuracy by 0.074, 0.061, and 0.049, and decreases the Logic Violation Rate by 7.50, 5.80, and 3.83 percentage points for Qwen3-8B, Qwen3-32B, and GPT-4o, respectively. Although the absolute reduction in LVR

Table 9
Cross-model generalization.

Absolute performance of NeuroSymbolic-Chain across sixteen backbones

Model	Extraction		Chain structure				Reliability	
	E-F1	R-F1	C-Acc	P-Sim	NED↓	RRMSE↓	LVR (%)↓	F-Con
Llama 2	0.654	0.582	0.415	0.688	0.285	0.264	8.95	0.692
Qwen3-8B	0.795	0.741	0.652	0.812	0.188	0.175	4.85	0.824
DeepSeek-R1-Distill-Qwen-7B	0.812	0.763	0.684	0.835	0.174	0.160	4.12	0.845
Llama 4 Maverick	0.845	0.801	0.732	0.865	0.145	0.138	3.25	0.880
DeepSeek-V3.2	0.875	0.838	0.782	0.906	0.123	0.116	2.24	0.913
Claude 3.7 Sonnet	0.865	0.826	0.773	0.896	0.132	0.123	2.58	0.904
GPT-4o	0.867	0.829	0.775	0.899	0.129	0.121	2.52	0.907
GLM-4.7	0.864	0.827	0.772	0.897	0.131	0.122	2.64	0.905
GLM-5	0.872	0.835	0.780	0.904	0.124	0.117	2.29	0.912
Doubao-1.5-Pro	0.861	0.823	0.768	0.895	0.134	0.124	2.71	0.903
Doubao-2.0-Pro	0.869	0.831	0.777	0.902	0.127	0.119	2.41	0.910
Kimi K2	0.883	0.846	0.790	0.911	0.118	0.111	2.03	0.919
Qwen3-32B	**0.878**	**0.842**	**0.785**	**0.908**	**0.121**	**0.114**	**2.15**	**0.915**
Claude 4.6 Sonnet	0.885	0.851	0.796	0.915	0.115	0.108	1.95	0.924
GPT-5	0.892	0.863	0.810	0.922	0.108	0.101	1.82	0.931
Qwen3-235B	0.884	0.849	0.792	0.913	0.116	0.109	1.98	0.922

Representative same-backbone comparison with GraphRAG

Backbone	GraphRAG		Ours		Δ	
	C-Acc	LVR (%)↓	C-Acc	LVR (%)↓	C-Acc	LVR (%)↓
Qwen3-8B	0.578	12.35	0.652	4.85	+0.074	−7.50
Qwen3-32B	0.724	7.95	0.785	2.15	+0.061	−5.80
GPT-4o	0.726	6.35	0.775	2.52	+0.049	−3.83

naturally narrows as the backbone model scales up, the relative reduction remains substantial (ranging from 60.3% to 73.0%). This trend demonstrates that while increased parametric capacity elevates the absolute performance ceiling, it does not replace the logic-preserving impact of explicit neuro-symbolic constraints.

Beyond these controlled comparisons, the broader evaluation across sixteen backbones (Table 9, upper block) provides a comprehensive context for calibration. Notably, even Llama 2, which exhibits the weakest reasoning capabilities within the evaluation matrix, limits the LVR to 8.95% under the proposed framework. Under the default configuration, Qwen3-32B achieves a Chain Accuracy of 0.785 and an LVR of 2.15%, performing near the upper bound of the evaluated backbones without requiring frontier-scale parameter capacity.

This controlled same-backbone comparison constitutes critical validation for the methodological novelty of the proposed framework: it confirms that the observed performance gains (e.g., Chain Accuracy improvements of 0.049–0.074 and LVR reductions of 3.83–7.50 percentage points) are directly attributable to the neuro-symbolic closed loop, thereby eliminating the alternative hypothesis that the enhancements are merely artifacts of a more powerful backbone model.

6.7. Time-controlled evaluation

Chronological dataset partitioning can confound temporal drift with intrinsic model capability. To mitigate this risk, we evaluated a reconstructed same-year slice centered on 2021, which represents the earliest post-2020 year exhibiting consistently high coverage in the archived raw sources. First, we merged individual reports into event units based on standardized locations, primary hazard types, and overlapping or adjacent time windows. Subsequently, we performed an event-level 80/20 split while freezing all hyperparameters from the main experimental configuration. This reconstructed control slice contains approximately N_{train} = 7.1k and N_{test} = 1.8k samples. Table 10 demonstrates that the performance ranking of the models remains consistent across all metrics. The proposed method continues to yield the highest scores for Entity F1, Relation F1, Chain Accuracy, Path Similarity, and Fact Consistency, while maintaining the lowest Normalized Edit Distance, Relative Root Mean Square Error, and Logic Violation Rate. These results indicate that the observed performance gains are unlikely to be driven solely by cross-year reporting drift.

6.8. Hyperparameter sensitivity and mechanism analysis

To investigate the behavior of the core neuro-symbolic components under various configurations and to empirically justify the parameter selection using the validation set (data from 2019 to 2022), this section presents a sensitivity analysis of the critical hyperparameters. Figs. 11 and 12 report the validation-set calibration curves for the temporal decay rate λ, the subgraph-matching threshold θ, and the Bayesian conflict-resolution parameters.

As illustrated by the left panel of Fig. 11, the overall performance of NeuroSymbolic-Chain is highly sensitive to the temporal decay rate λ within the time-aware forget gate $\eta(\Delta t)$. The adopted temporal decay setting (λ = 0.18) yields the optimal balance between retaining long-range causal preconditions and adapting to abrupt disaster states. Deviating from this operating point results in notable performance degradation. When λ is excessively low (e.g., 0.06), the system becomes prone to state stagnation, where obsolete historical states fail to decay in a timely manner. This stagnation amplifies conflicts between historical and recent evidence during the generation phase, thereby increasing the LVR to 2.74%. Conversely, when λ is excessively high (e.g., 0.30), the model discards key prerequisites prematurely, such as early soil saturation states. This premature forgetting disrupts the structural integrity of long-chain evolution, subsequently reducing Chain Accuracy and Path Similarity to 0.763 and 0.884, respectively. This trend demonstrates that nonlinear temporal constraints are not merely auxiliary details; rather, they are essential for filtering purely statistical correlations and establishing physically grounded evolutionary explanations.

As illustrated by the right panel of Fig. 11, within the topology-aware dynamic graph retrieval module, the subgraph matching threshold θ governs the balance between generative strictness and retrieval tolerance. At θ = 0.74, the system achieves the most robust trade-off between filtering causal hallucinations and retaining valid risk paths, yielding a Chain Accuracy of 0.778 on the validation set. A lower threshold (e.g., θ = 0.62) retrieves numerous locally isomorphic subgraphs containing structural noise and spurious associations. This dilutes the supervisory signal imposed by the Principles of Narrative Invariance and consequently elevates the LVR to 2.46%. Conversely, an overly strict setting (e.g., θ = 0.80) suppresses noisy motifs but

Table 10
Time-controlled evaluation.

Model	Extraction		Chain structure				Reliability	
	E-F1	R-F1	C-Acc	P-Sim	NED↓	RRMSE↓	LVR (%)↓	F-Con
Pure LLM	0.748	0.693	0.618	0.656	0.350	0.307	16.37	0.712
Vanilla RAG	0.731	0.704	0.584	0.639	0.372	0.325	12.84	0.764
ChronoRAG	0.781	0.731	0.626	0.699	0.291	0.246	7.28	0.816
GraphRAG	0.812	0.756	0.681	0.817	0.206	0.172	8.64	0.854
RAPTOR	0.806	0.761	0.689	0.821	0.201	0.168	6.89	0.861
HippoRAG	0.821	0.773	0.697	0.836	0.191	0.159	5.94	0.873
LightRAG	0.828	0.769	0.704	0.851	0.181	0.146	5.41	0.869
DyG-RAG	0.825	0.771	0.712	0.848	0.185	0.149	4.97	0.871
OGPE-only	0.798	0.752	0.661	0.754	0.237	0.197	5.33	0.841
Ours	**0.862****	**0.821****	**0.741****	**0.881****	**0.143****	**0.133****	**2.41****	**0.899****

** Indicates Holm–Bonferroni-corrected $p < 0.001$ compared with the strongest competing baseline for the corresponding metric. The same paired testing and multiple-comparison correction protocol described in Section 6.1.6 was applied.

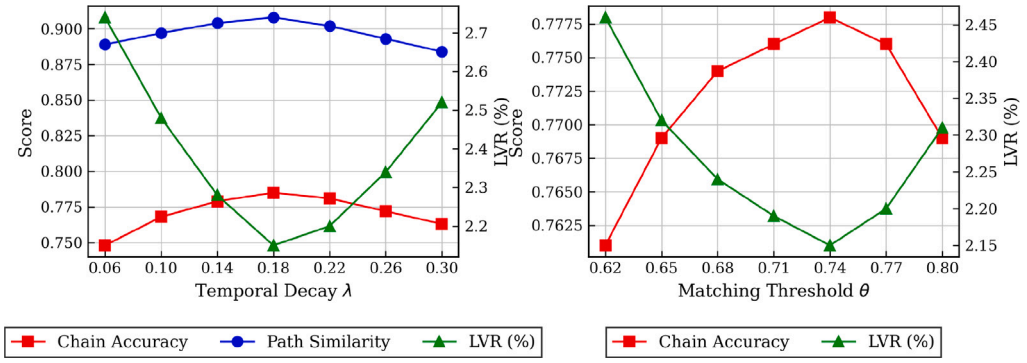


Fig. 11. Sensitivity to λ and θ. The selected operating points are $\lambda = 0.18$ and $\theta = 0.74$.

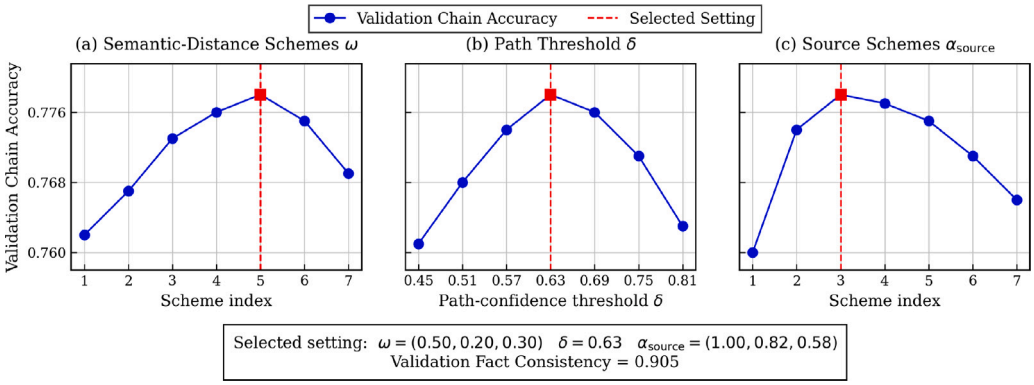


Fig. 12. Bayesian conflict-resolution calibration. The curves report validation Chain Accuracy, and the selected setting yields a Fact Consistency of 0.905.

inadvertently eliminates valid isomorphic subgraphs that remain physically sound despite minor lexical variations. This reduction in structural diversity causes the Chain Accuracy to drop to 0.769. Therefore, the objective of tuning θ is not to maximize strictness in isolation, but rather to maintain an effective balance between logical rigidity and chain completeness.

Fig. 12 further shows that the Bayesian conflict-resolution module is most stable when the adopted setting lies on a broad high-performing plateau rather than at an isolated spike. This behavior is important because the module must arbitrate heterogeneous evidence without over-trusting noisy social signals or over-pruning valid minority reports. Notably, the adopted ω-scheme and α_{source}-scheme are selected at different scheme indices rather than collapsing to a single central index, which reduces the risk that the observed calibration is merely a midpoint artifact of the search grid. The selected configuration, namely $\alpha_{source} = (1.00, 0.82, 0.58)$, $\omega = (0.50, 0.20, 0.30)$, and $\delta = 0.63$, achieves the highest validation Fact Consistency score of 0.905 while preserving strong chain-level accuracy. Stable performance in noisy disaster-semantic space therefore depends on coordinated calibration among retrieval tolerance, logical constraints, and multi-source evidence arbitration thresholds.

6.9. Logical consistency analysis

Granular Composition of Logic Violations. To avoid repeatedly reporting conflicting absolute LVR totals across tables, Table 11 normalizes the three exhaustive violation categories within the total LVR of three representative systems, while the final row provides the absolute LVR values aligned with Table 5 for reference.

A granular analysis reveals that, within the representative systems in Table 11, C_{temp} and C_{causal} constitute the largest proportions,

Table 11
Logic-violation composition.

Violation type	Example	Pure LLM (%)	GraphRAG (%)	Ours (%)
C_{temp}	$t_{\text{cause}} > t_{\text{effect}}$	35.9	36.9	30.7
C_{causal}	$A \to B \to A$	41.3	40.4	40.1
$C_{\text{intensity}}$	$S_{\text{text}} \gg S_{\text{phy}}$	22.8	22.7	29.2
Total LVR	–	**14.82**	**7.95**	**2.15**

Note: The upper three rows show the normalized internal composition, summing to 100%, of logic violations for each model. C_{temp}, C_{causal}, and $C_{\text{intensity}}$ denote temporal inversion, causal cycle, and intensity mismatch, respectively. The bottom row reports the absolute Total Logic Violation Rate (LVR).

whereas $C_{\text{intensity}}$ is relatively minor. Specifically, C_{temp} denotes temporal inversion, C_{causal} denotes causal cycle, and $C_{\text{intensity}}$ denotes intensity mismatch. For instance, the model frequently generates reverse causal relations, such as floods triggering rainstorms. Combined with the absolute LVR values aligned with Table 5, this pattern demonstrates that formalizing meteorological physical principles into reusable constraint predicates, and subsequently enforcing these constraints through guided generation and consistency checking, effectively reduces the overall volume of physically inconsistent outputs, rather than merely shifting violations from one audited category to another.

6.10. Ablation study

Overall Ablation Results. To isolate and validate the methodological novelty of the core mechanisms within the proposed framework, we conducted a rigorous ablation study (Table 12). This analysis demonstrates that the proposed modules function as fundamental drivers of reasoning fidelity rather than superficial additions. Specifically, the first four variants omit major components of the framework, whereas the final two variants retain the dynamic update pipeline but replace specific internal mechanisms with simplified baselines to strictly isolate their individual contributions. To further address the runtime cost of each component, we additionally report the isolated per-sample E2E latency and its change relative to the full NeuroSymbolic-Chain framework.

The latency columns in Table 12 are intended to quantify the runtime cost associated with each architectural component. This comparison is important because the proposed framework improves logical consistency and factual grounding by introducing ontology-guided extraction, topology-aware graph retrieval, physical constraints, dynamic updating, and Bayesian conflict resolution. Therefore, the ablation results should be interpreted not only in terms of accuracy and reliability gains, but also in terms of the additional or reduced E2E runtime introduced by each component.

Component-wise Interpretation. Our analysis reveals that each component exerts a distinct, orthogonal influence within the neuro-symbolic closed loop, validating their collective necessity:

1. **Logical Consistency via Constraints:** The constraint mechanism is pivotal for maintaining logical coherence. Ablating PNI-based constraints triggers a sharp LVR increase (+2.83 pp), isolating unconstrained generation as the primary driver of hallucinatory content. Specifically, the removal of the physical constraints led to a substantial increase in the LVR from 2.15% to 4.98%, thereby confirming that physical-rule constraints serve as the primary mechanism for mitigating logical inconsistencies.
2. **Semantic Grounding via OGPE:** OGPE serves as the core source of semantic grounding. Excluding the ontology-guided module precipitates the most severe degradation in Entity F1 (−0.072), demonstrating that without rigid schema constraints, the model succumbs to severe semantic drift amidst fragmented text streams. Specifically, the removal of OGPE resulted in a decline in the Entity F1 score from 0.878 to 0.806, thereby demonstrating its independent contribution to semantic normalization.
3. **Reasoning Depth via GraphRAG:** GraphRAG extends inferential capacity. While its removal impacts entity extraction minimally, the loss of graph retrieval severely compromises factual consistency, underscoring the critical role of topological retrieval in evidence cross-validation.
4. **Time-aware Forget Gate Supports Chain-level Temporal Consistency.** Replacing the adaptive temporal decay with a time-independent state-mixing baseline significantly decreases Chain Accuracy and elevates the LVR, while the Entity F1 score experiences only marginal variations. This indicates that the primary role of the mechanism lies in maintaining chronological consistency at the chain level, rather than in enhancing local extraction fidelity.
5. **Bayesian Conflict Resolution Supports Factual Consistency under Heterogeneous Evidence.** Replacing source-aware conflict arbitration with a uniform-source weighting baseline significantly decreases Fact Consistency and elevates the LVR. This finding indicates that source-aware conflict arbitration contributes to maintaining factual consistency in scenarios of heterogeneous evidence disagreement.

7. Case study

To validate the reasoning capabilities of the model under extreme real-world conditions, we selected the July 2023 extreme rainstorm in Beijing as a primary case study. This event exemplifies severe data fragmentation. Initially, social media chatter fixated on urban waterlogging and traffic congestion, obscuring critical signals of geological instability in remote mountainous regions. We benchmarked our approach against a traditional vector retrieval baseline (Vanilla RAG), with the comparative reasoning process detailed in Fig. 13.

As the analysis shows, the system identified latent risks that the baseline did not recover:

- **Implicit Reasoning across the Semantic Gap.** Constrained by surface-level semantic matching, traditional vector retrieval offered only generic advice regarding drainage maintenance (MRR: 0.62). By contrast, NeuroSymbolic-Chain moved beyond surface urban-flooding signals. By computing motif fingerprints for subgraph structures, it retrieved highly isomorphic historical precedents from the 21 July 2012 rainstorm in Beijing [69].
- **Deep Reconstruction of Physical Mechanisms.** Leveraging historical motifs, the system reconstructed the complete physical evolution sequence: Continuous Heavy Rainfall → River Level Surge → Lateral Erosion → Riparian Subgrade Hollowing.
- **Targeted Risk Indication.** Even absent direct reports of road failure, the model identified the risk of Subgrade Hollow & Collapse and highlighted riparian bridges for expert inspection. This text-driven indication aligned with subsequent road damage assessments and shows how historical motif matching can recover chain-level warning evidence that is difficult to obtain from surface lexical retrieval alone. In operational use, such evidence is intended to support expert triage, together with specialized hydrodynamic simulators, rather than to replace operational judgment.

This case highlights two practical applications of the NeuroSymbolic-Chain framework under conditions of fragmented reporting: identifying latent risks prior to the emergence of explicit failure reports, and providing mechanistic explanations that connect upstream rainfall with downstream infrastructure instability. Combined with the shared event-level benchmark reported above, these findings demonstrate that the proposed pipeline effectively supports situational assessment in text-intensive disaster scenarios. Residual false positives, over-correction, and their quantitative failure boundaries are discussed in Section 8.1.

Table 12
Ablation results with efficiency trade-off.

Model variant	Extraction			Chain	Reliability			Efficiency	
	E-F1	ΔE-F1	R-F1	C-Acc	LVR (%)↓	ΔLVR (pp)	F-Con	E2E Lat. (s)↓	ΔLat. (s)
Ours	**0.878**	–	**0.842**	**0.785**	**2.15**	–	**0.915**	**1.96 ± 0.35**	–
w/o OGPE	0.806	−0.072	0.775	0.702	4.48	+2.33	0.858	1.72 ± 0.28	−0.24
w/o GraphRAG	0.823	−0.055	0.788	0.724	3.42	+1.27	0.876	1.41 ± 0.22	−0.55
w/o Constraints	0.855	−0.023	0.812	0.745	4.98	+2.83	0.885	1.86 ± 0.31	−0.10
w/o Dyn. Update	0.846	−0.032	0.805	0.738	3.07	+0.92	0.890	1.79 ± 0.30	−0.17
TAFG baseline	0.873	−0.005	0.836	0.736	4.18	+2.03	0.895	1.90 ± 0.33	−0.06
BCR baseline	0.865	−0.013	0.824	0.755	3.85	+1.70	0.852	1.84 ± 0.30	−0.12

Note: Delta columns are measured against the full NeuroSymbolic-Chain model. ΔLVR is reported in percentage points (pp), and positive values indicate degraded logical consistency relative to the full system. E2E latency is measured using the same isolated per-sample end-to-end protocol as Table 6. ΔLatency is computed relative to the full NeuroSymbolic-Chain framework; negative values indicate reduced runtime, whereas positive values indicate additional runtime. Component-wise degradation was evaluated on the same held-out test split as the full model. When instance-level paired outputs are available, paired significance testing follows the protocol described in Section 6.1.6. Unlike the main benchmark comparisons in Table 5, this table is intended to isolate the functional contribution and efficiency trade-off of each component rather than to serve as a separate multiple-comparison significance table.
Abbreviations: TAFG baseline denotes Time-aware Forget Gate with a time-independent mixing baseline; BCR baseline denotes Bayesian Conflict Resolution with a uniform-source weighting baseline.

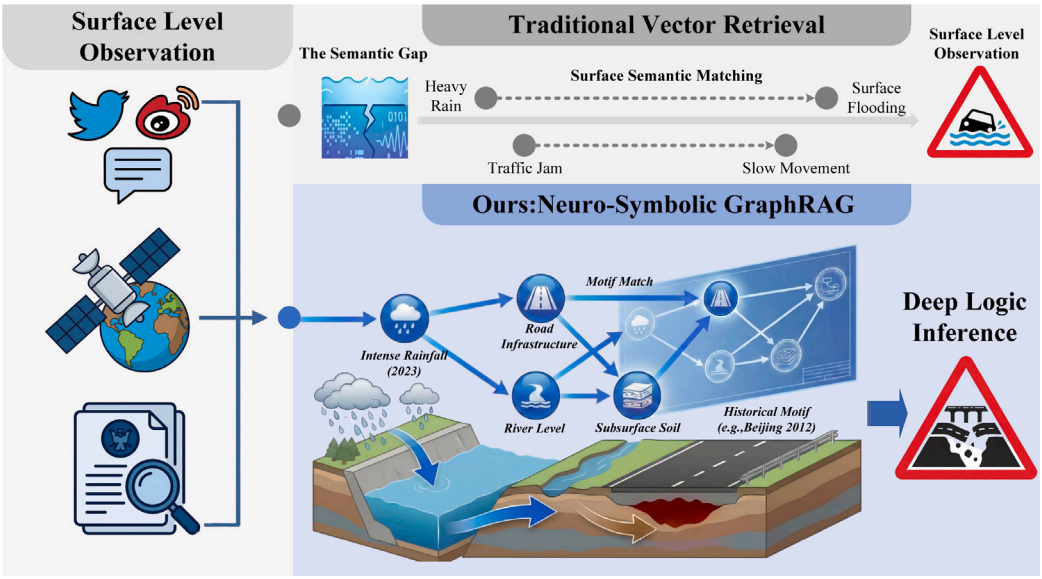


Fig. 13. Beijing rainstorm case study. Neuro-symbolic retrieval recovers subgrade hollowing that vanilla vector retrieval misses.

8. Discussion

We introduced NeuroSymbolic-Chain, a neuro-symbolic framework synergizing Large Language Models (LLMs) with Dynamic Causal Knowledge Graphs to resolve the complexities of parsing meteorological disaster chains from heterogeneous text streams. Addressing the semantic gaps and logical hallucinations endemic to conventional long-chain reasoning, our core contributions are threefold:

1. A Neuro-Symbolic Closed Loop with a Task-Specific Instantiation of Narrative Invariance. We instantiate PNI as a set of executable disaster-chain constraint predicates and integrate them into structured prompting, candidate validation, and graph write-back. Rather than claiming novelty for temporal or logical constraints in the abstract, our contribution lies in operationalizing these general constraint ideas for meteorological disaster narratives and coupling them with the retrieval-validation-write-back pipeline. This design reduces logical hallucinations in generated disaster narratives.

2. Topology-Aware GraphRAG for Structural Risk Retrieval. Departing from shallow semantic similarity, we engineered a retrieval stage driven by subgraph isomorphism and topological serialization. Coupled with the downstream constraint layer, this stage captures latent topological structures within historical motifs and supports the recovery of concealed cascading risks (e.g., Rainfall → Subgrade Failure).

3. Ontology-Guided Prompt Engineering (OGPE) & Dual-Stream Textual Updates. We integrated domain ontology as a cognitive prior with a Bayesian conflict resolution mechanism. This strategy resolves data fragmentation and narrative conflict, ensuring rigorous logical and factual consistency throughout the dynamic evolution of the knowledge graph.

Extensive evaluations on our Multi-source Disaster Semantic Corpus (MDSC) — spanning 100,000+ records (2000–2025) — demonstrate strong performance of the framework. In particular, for logical reliability, the framework reduces the LVR from 4.52% for DyG-RAG to 2.15%, a relative reduction of 52.4%; meanwhile, compared with the strongest competing results on the corresponding metrics, Entity F1 improves by 3.8%, Chain Accuracy by 5.9%, Path Similarity by 4.1%, while NED and RRMSE decrease by 27.1% and 18.6%, respectively. The case study of the July 2023 extreme rainstorm in Beijing further shows that the framework can surface structural risks (e.g., subgrade hollowing) from fragmented early reports, supporting complex risk interpretation in scenarios where text evidence arrives earlier than a complete chain narrative. Crucially, the methodological novelty of NeuroSymbolic-Chain does not arise from the deployment of temporal constraints, ontology prompting, or graph retrieval as isolated techniques. Instead,

it lies in the systematic integration of these mechanisms into a verifiable neuro-symbolic closed loop tailored for meteorological disaster text reasoning. A candidate chain within the proposed framework must successfully undergo ontology normalization, topological motif retrieval, physical-rule filtering, and Bayesian conflict arbitration before achieving final validation. As demonstrated through the mechanism-isolation and same-backbone analyses, this tightly coupled architecture constitutes the primary factor for fundamentally mitigating logical violations, thereby distinguishing the proposed framework from the simple pipeline concatenations of existing RAG methods. Consequently, the proposed framework serves as a text-driven complement to specialized meteorological and hydrodynamic simulators.

The category-level analysis in Table 7 complements these aggregate benchmarks by elucidating cross-category variations relative to DyG-RAG across the primary hazard types within the profiled subset. Sudden-onset categories demonstrate narrower improvement margins because their causal chains are structurally shorter and denser. By contrast, the long-latency drought subset remains the clearest stress case and the primary source of relative performance gains, thereby providing a focal point for future refinement.

8.1. Limitations and failure modes

Residual errors mainly take two forms: under-supported acceptance and over-pruning of unconventional chains. In scenarios characterized by sparse evidence and delayed conflict signals, superficially coherent narratives can evade correction mechanisms, as demonstrated by the failure case concerning the hydrological propagation lag discussed in Section 6.5. Conversely, under the influence of novel mechanisms driven by climate change, valid yet unconventional causal chains may be erroneously pruned, with Drought-Flood Abrupt Alternation serving as a definitive boundary case within the subset of novel mechanisms. Under the FC protocol, a score of 0.915 still means that approximately 8.5% of atomic assertions remain unsupported or contradicted after correction, and under the most adverse stress condition, namely PNI-consistent narratives combined with sparse official evidence, the false-acceptance rate rises to 28.6%. These results characterize the closed loop as an error-reduction mechanism with clear operating boundaries, not a complete safeguard against hallucination.

These boundaries have several practical consequences. Performance declines on structurally novel hazard chains and longer multi-hop cascades, and cross-regional transfer under substantial domain shift requires recalibration. On the climate-shifted novel-mechanism subset, the recall of valid unconventional paths falls to 0.294, while the false-pruning rate rises to 33.5%. Bayesian conflict resolution reduces routine contradictions, but it does not fully resolve fundamental disagreements among authoritative agencies or structurally coherent misinformation under sparse official evidence. In addition, bounded DFS-based implicit deduction can over-infer when explicit textual anchors are absent, with a residual false-positive rate of 26.2% versus 2.3% for purely explicit textual trajectories. Furthermore, the processing of high-velocity bursts of social media data imposes substantial computational demands. Specifically, under the extreme 8× burst setting, the end-to-end latency surges to 54.2 s, accompanied by severe queue buildup, a timeout or drop rate of 22.8%, and a decline in the triggered feedback ratio to 6.3%. This observation demonstrates that system overload is driven primarily by stream-level queue pressure and load shedding, rather than by computational exhaustion within the Bayesian feedback loop. Finally, because the framework operates on observational text, latent confounders cannot be eliminated completely. The system is therefore positioned as a text-based decision-support component that proposes physically plausible candidate chains, rather than as a standalone causal identification or physical forecasting system.

8.2. Ethical considerations and responsible deployment

Because disaster-chain analysis may influence triage and resource allocation, NeuroSymbolic-Chain is positioned as a decision-support tool rather than an autonomous dispatch system. The most immediate fairness risk arises from uneven regional coverage in MDSC: underrepresented areas may receive weaker motif support and less stable conflict calibration. Region-stratified error audits, minimum evidence-coverage checks, and periodic supplementation with local official bulletins are therefore required before operational use; outputs from sparsely represented regions are treated as lower-confidence candidates. In addition, any predicted chain that could affect evacuation, road closure, emergency dispatch, or cross-agency resource allocation remains subject to qualified analyst review before field action is taken.

Responsible use also requires transparent prioritization and explicit uncertainty handling. Operational users must be able to inspect the matched historical motif, the evidence supporting each propagated edge, and the source-authority weighting used in conflict resolution. Section 8.1 shows that residual uncertainty remains material, including approximately 8.5% unsupported or contradicted atomic assertions under the FC protocol, a 28.6% false-acceptance rate under sparse-official-evidence stress, and a 26.2% false-positive rate for DFS-dependent trajectories. Cases meeting these conditions are treated as review-required candidates pending external corroboration rather than as actionable alerts.

8.3. Future work

While NeuroSymbolic-Chain significantly advances logical interpretability in disaster analysis, several avenues warrant further exploration:

1. **Cross-Modal Perception & Joint Reasoning.** Moving beyond text, we aim to integrate satellite imagery and radar telemetry to construct a vision-language aligned Disaster Knowledge Graph [70], thereby enhancing the spatiotemporal resolution of physical state perception.
2. **Evolution toward Agentic Counterfactual Reasoning.** We envision transforming the system from a passive analytical tool into an agentic system. By incorporating counterfactual reasoning, the model will execute "what-if" simulations, providing city managers with prospective, dynamic risk assessments.
3. **Uncertainty Quantification and Hybrid Modeling.** Future research could integrate probabilistic modules, such as Gaussian process regression [71], into the current chain reasoning framework. Such integration would yield calibrated estimates of uncertainty and confidence rankings for propagated risk trajectories, thereby enhancing the transparency of decision-making in high-risk settings.

Ultimately, this work establishes an interpretable decision-support layer for disaster management. Its outputs are intended for analyst-mediated situational assessment under explicit uncertainty disclosure, not for autonomous resource-allocation decisions, while remaining compatible with specialized physical simulators used in operational meteorology, hydrology, and urban infrastructure planning.

CRediT authorship contribution statement

Nengfu Xie: Writing – review & editing, Writing – original draft, Visualization, Software, Project administration. **JingChao Fan:** Project administration, Investigation, Funding acquisition, Conceptualization. **Lihua Jiang:** Resources, Project administration, Methodology, Investigation, Funding acquisition. **Xin Wang:** Writing – review & editing, Writing – original draft, Project administration, Investigation, Data curation. **Xiaoli Wang:** Software, Resources, Project administration,

Methodology, Conceptualization. **Hailong Liu:** Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Huoguo Zheng:** Software, Resources, Investigation, Funding acquisition. **Zhibo Meng:** Writing – review & editing, Writing – original draft, Visualization, Validation. **Hao Wu:** Visualization, Validation, Software, Resources. **Xiaoxiao Li:** Resources, Project administration, Methodology, Investigation. **Qianjie Lv:** Validation, Software, Investigation, Formal analysis. **Yuyu Ren:** Investigation, Formal analysis, Data curation, Conceptualization. **Huanping Wu:** Project administration, Methodology, Investigation, Formal analysis, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Core reproducibility artifacts

A.1. Final YAML configuration file

```yaml
backbone:
  model: Qwen3-32B
  token_budget: 4096
  temperature: 0.5
  top_p: 0.9
representation:
  implicit_state_dim: 768
sdcr:
  eps_dbscan: 0.42
  min_samples: 4
temporal:
  # lambda in the time-aware forget gate
  forget_gate_lambda: 0.18
  delta_min_hours: 1.5
matching:
  # omega in the SDCR distance
  omega:
    semantic: 0.50
    temporal_gap: 0.20
    topology_neighbor: 0.30
  theta: 0.74
path_scoring:
  # alpha and delta in Score(P)
  time_gap_alpha: 0.80
  path_confidence_delta: 0.63
conflict_resolution:
  beta: 0.12
  epsilon: 1.0e-5
  source_authority:
    government: 1.00
    news: 0.82
    social: 0.58
retrieval:
  k_motif: 12
search:
  bounded_dfs_max_depth: 5
feedback:
  max_rounds: 3
```

A.2. Complete ontology design of P_{Ω}

See Table A.13.

A.3. Complete normalized prompt templates

Extraction-stage prompt template

```
<system_role>
You are a meteorological-disaster event extractor.
    Your only task is to produce structured candidate-
    edge JSON objects grounded in the input text and
    the provided P\Omega ontology and runtime schema;
    each object contains the subject-relation-object
    core together with the runtime fields required for
    write-back. Do not produce free-form conclusions
    outside the defined schema boundary.
</system_role>

<task_objective>
1. Identify event nodes, state nodes, and monitoring
    signals in the text.
2. Extract subject-relation-object candidates and fill
     the core fields required for graph write-back.
3. For candidates that remain uncertain but should
    still be retained, do not repair them by invention;
     return only normalized candidates together with
    the temporal, spatial, and intensity fields
    required for downstream symbolic validation.
</task_objective>

<input_contract>
<source_metadata>
- source_id: <string>
- source_type: <government|news|social>
- normalized_time: <hour-bucketed datetime or null>
- normalized_location: <canonical location string or
    null>
</source_metadata>
<raw_evidence>
- raw_text: <document span>
- optional_context: <adjacent sentences or null>
</raw_evidence>
<ontology_reference>
- allowed_entity_types \in {Meteorological Driver,
    Hydrological Response, Environmental Condition,
    Geological Hazard, Urban Waterlogging, Coastal
    Marine Hazard, Cryospheric Visibility Hazard,
    Agricultural Ecological Impact, Infrastructure
    Failure, Transportation Disruption, Lifeline
    Outage, Community Vulnerability, Population
    Exposure, Emergency Action, Monitoring Signal,
    Damage Report, Resource Status, Recovery Status}
- allowed_relations \in {causes, exacerbates, precedes
    , co_occurs_with, affects, blocks, damages,
    weakens, exposes, mitigates, supported_by,
    updates_state_of}
- logical-only projections: ‘located_in ->
    location_name/location_level‘, ‘reported_by ->
    evidence_source‘
- candidate_fields and validator_backfill_fields
    follow Table A3
</ontology_reference>
</input_contract>

<execution_procedure>
1. Normalize time, location, and intensity mentions
    first, and floor comparable times to hourly
    buckets; use ‘null‘ when they cannot be resolved.
```

Table A.13
Ontology design of P_{Ω}.

Upper-domain/Component	Item	Definition/Scope	Closed vocabulary/Domain-range
Layer: Runtime schema			
Candidate fields I	`event_id`, `entity_type`, `trigger_text`, `time_start`, `time_end`, `location_name`, `location_level`, `intensity_level`	Basic node typing, temporal, spatial, and intensity fields required for candidate nodes or candidate edges	Fixed JSON keys preserved as listed; times are floored to hourly buckets at runtime
Candidate fields II	`evidence_source`, `confidence`, `status`	Evidence-source typing, fused confidence, and lifecycle status	Fixed JSON keys preserved as listed; `evidence_source` is the runtime projection of logical `reported_by`
Candidate auxiliary fields	`affected_object`, `upstream_event`, `notes`	Auxiliary linkage and note fields for candidate edges or chains	Fixed JSON keys; may be empty arrays or empty strings
Validator back-fill fields	`screening_result`, `violation_tags`	Screening outputs appended by the external deterministic symbolic validator after candidate checking	Not produced directly by the model
Layer: Canonical entity class			
Meteorological forcing	Meteorological Driver	Direct meteorological forcing such as heavy rainfall, strong wind, cold-air outbreaks, or heat stress	Rainstorm; torrential rain; extreme rainfall; short-duration heavy rainfall; continuous rainfall; severe convection; thunderstorm gale; tornado; lightning; hail; typhoon; tropical storm; severe tropical storm; cold wave; low temperature; heatwave; drought; dry-hot wind; sandstorm; blowing dust
Environmental and process state	Hydrological Response	River rise, flood evolution, runoff accumulation, and related hydrological responses	River surge; medium-small river flood; flash flood; flood; reservoir rise; runoff accumulation; snowmelt flood
Environmental and process state	Environmental Condition	Background states that modulate hazard evolution, including moisture, icing, heat, and visibility conditions	Soil saturation; bank erosion; drainage deterioration; snow accumulation; icy condition; visibility deterioration; heat-dry surface condition
Hazard and infrastructure impact	Geological Hazard	Cascading geophysical hazards triggered or amplified by meteorological forcing	Landslide; collapse; debris flow; ground instability
Hazard and infrastructure impact	Urban Waterlogging	Urban inundation processes affecting roads, tunnels, and underground spaces	Urban inundation; roadway ponding; underground backflow; underpass flooding; tunnel flooding
Hazard and infrastructure impact	Coastal Marine Hazard	Coastal and estuarine hazards induced by storms and marine backwater effects	Storm surge; coastal inundation; estuary backwater; seawater intrusion
Hazard and infrastructure impact	Cryospheric Visibility Hazard	Snow, icing, fog, and haze hazards that disrupt mobility and safety	Blizzard; snow disaster; freezing rain; glaze icing; rime icing; icy road; dense fog; haze; blowing snow
Hazard and infrastructure impact	Agricultural Ecological Impact	Agricultural and ecological impacts driven by meteorological extremes	Cropland waterlogging; crop drought stress; low-temperature crop damage; frost damage; forest fire; grassland fire; pest outbreak
Hazard and infrastructure impact	Infrastructure Failure	Physical damage or instability of engineered assets under disaster stress	Subgrade hollowing; bridge damage; embankment failure; drainage-pipe rupture; dam breach; power-line icing damage
Hazard and infrastructure impact	Transportation Disruption	Mobility and logistics disruption caused by meteorological hazards or cascading impacts	Road closure; traffic paralysis; rail interruption; flight disruption; logistics blockage
Hazard and infrastructure impact	Lifeline Outage	Interruptions to essential utility and service networks	Power outage; communication interruption; water-supply interruption; heating interruption
Exposure and vulnerability	Community Vulnerability	Receptors whose geographic or structural conditions increase hazard susceptibility	Old community; low-lying district; mountain-front village; coastal settlement
Exposure and vulnerability	Population Exposure	Population-level exposure, stranding, and sheltering pressure	Population stranding; evacuation demand; shelter pressure; tourist exposure
Response and recovery	Emergency Action	Emergency measures deployed to suppress, divert, or respond to risk	Drainage dispatch; evacuation order; traffic control; pumping deployment; emergency rescue; reservoir discharge regulation
Observation and evidence	Monitoring Signal	Observational or reported signals used as evidence anchors	Rain-gauge anomaly; water-level anomaly; displacement alert; radar echo intensification; wind-speed anomaly; icing observation; social-media help signal
Hazard and infrastructure impact	Damage Report	Outcome descriptors summarizing human, economic, and asset losses	Economic loss; casualty report; housing damage; crop loss; infrastructure loss
Response and recovery	Resource Status	Availability or depletion of key response resources	Emergency supplies; pumping capacity; transport capacity; shelter capacity

(*continued on next page*)

Table A.13 (*continued*).

Response and recovery	Recovery Status	Restoration and return-to-service progress after impact	Water recession; road reopening; power recovery; production resumption
Layer: Relation type			
Relation inventory	`causes`	Direct causal triggering	Canonical event/state node → canonical event/state node
Relation inventory	`exacerbates`	Aggravates an existing hazard or outcome	Canonical event/state node → canonical event/state node
Relation inventory	`precedes`	Explicit temporal precedence	Canonical event/state node → canonical event/state node
Relation inventory	`co_occurs_with`	Temporal co-occurrence without direct causal encoding	Canonical event/state node → canonical event/state node
Relation inventory	`located_in`	Spatial-assignment logical relation	Projected in the prompt-deployed runtime to `location_name` / `location_level`; not emitted as a standalone edge
Relation inventory	`affects`	Impacts infrastructure, population, resource, or regional state	Canonical event/state node → {Infrastructure Failure, Lifeline Outage, Population Exposure, Resource Status, Damage Report, Community Vulnerability}
Relation inventory	`blocks`	Blocks mobility or logistics	Canonical event/state node → Transportation Disruption
Relation inventory	`damages`	Damages infrastructure or amplifies losses	Canonical event/state node → {Infrastructure Failure, Damage Report}
Relation inventory	`weakens`	Progressively weakens structural stability	Environmental Condition → Infrastructure Failure
Relation inventory	`exposes`	Places a population or receptor at risk	Canonical forcing/hazard/state node → {Population Exposure, Community Vulnerability}
Relation inventory	`mitigates`	Suppresses or alleviates a risk	Emergency Action → canonical event/state node
Relation inventory	`supported_by`	Supported by an observation or text evidence anchor	Canonical event/state node → Monitoring Signal
Relation inventory	`reported_by`	Evidence-source logical relation	Projected in the prompt-deployed runtime to `evidence_source`; not emitted as a standalone edge
Relation inventory	`updates_state_of`	Triggers a state update of another node	Canonical event/action node → {Resource Status, Recovery Status, Lifeline Outage, Population Exposure, Infrastructure Failure}

```
2. Extract only nodes and relations that can be
     anchored by ‘trigger_text‘ or an adjacent
     supporting sentence.
3. Both ‘subject‘ and ‘object‘ must map to canonical
     types in P\Omega; if a mention cannot be mapped,
     drop the candidate rather than inventing a new type
     .
4. Fill ‘confidence‘, ‘status‘, ‘notes‘, and the
     temporal, spatial, and intensity fields needed by
     the downstream validator for every candidate;
     serialize spatial and source bindings through ‘
     location_*‘ and ‘evidence_source‘ rather than
     emitting standalone ‘located_in‘ / ‘reported_by‘
     edges.
5. Do not perform algebraic rule checking inside the
     model; return JSON candidates only, with no
     explanatory prose, tables, or Markdown.
</execution_procedure>

<grounding_policy>
- Do not invent events, locations, times, intensities,
     or causal directions that are not supported by the
     text or normalized context.
- If the text supports co-occurrence but not causality,
     prefer ‘co_occurs_with‘; if even co-occurrence is
     weak, drop the candidate.
- If temporal order, spatial connectivity, or
     intensity evidence is insufficient, lower
     confidence or drop the candidate; do not fabricate
     ‘screening_result‘ or ‘violation_tags‘.
- Do not emit standalone ‘located_in‘ or ‘reported_by‘
     edges; write spatial and source bindings into the
     attribute fields instead.
- If no qualified candidates are found, return ‘[]‘.
</grounding_policy>

<symbolic_validation_reference>
The following checks are executed after generation by
     an external deterministic symbolic validator and
     are included only to define the admissible output
     boundary. The runtime model is not responsible for
     algebraic checking or for filling ‘
     screening_result‘ / ‘violation_tags‘.
1. ‘Ctemp‘: the validator checks whether ‘t(cause) + \
     delta_{min} \leq t(effect)‘.
2. ‘Ccausal‘: the validator checks geographical
     adjacency, physical connectivity, and explicit
     cycles.
3. ‘Cintensity‘: the validator checks whether
     downstream consequence intensity is compatible
     with upstream energy input.
4. Out-of-schema: if the entity type or relation type
     falls outside P\Omega, the candidate is not
     written back to the graph.
</symbolic_validation_reference>

<output_contract>
Return a JSON array only. Do not output Markdown, prose
     , or extra keys. Each candidate item must satisfy:
{
  "subject": {"event_id": "...", "label": "...", "
     entity_type": "..."},
```

```
  "relation": "...",
  "object": {"event_id": "...", "label": "...", "
     entity_type": "..."},
  "trigger_text": "...",
  "time_start": "...",
  "time_end": null,
  "location_name": "...",
  "location_level": "...",
  "intensity_level": "...",
  "evidence_source": "...",
  "confidence": 0.0,
  "status": "observed|inferred|resolved",
  "affected_object": [],
  "upstream_event": [],
  "notes": ""
}
Formatting constraints:
- use `null` or `[]` for unknown optional fields
- do not omit required keys
- do not add schema-external keys (equivalent to `
     additionalProperties=false`)
- each object represents one candidate edge
- when times are comparable, sort by `time_start`;
     otherwise preserve source order
Post-processing note:
- the object above is the candidate schema
- the external symbolic validator appends `
     screening_result` and `violation_tags` after
     generation
</output_contract>
```

Generation-stage prompt template

```
<system_role>
You are a disaster-chain analyst. Infer latent risk
     chains from the serialized subgraph, historical
     motifs, and structured constraint context, and
     return candidate chains in structured JSON.
</system_role>

<task_objective>
1. Infer plausible latent risk chains from the provided
     `Gsub`.
2. For each candidate chain, provide the node sequence,
     edge relations, evidence anchors, and confidence.
3. If a chain depends on a missing bridge or
     insufficient evidence, do not patch it by
     inventing intermediate nodes; return only
     defensible candidates together with the fields
     required for downstream symbolic validation.
</task_objective>

<input_contract>
- chain_context: <serialized Gsub>
- motif_paths: <historical motifs>
- node_inventory: <node ids, types, timestamps,
     locations, status>
- edge_relation_inventory: {causes, exacerbates,
     precedes, co_occurs_with, affects, blocks,
     damages, weakens, exposes, mitigates,
     supported_by, updates_state_of}
- evidence_summary: <supporting spans or null>
- horizon: <forecast horizon or null>
</input_contract>

<reasoning_procedure>
1. Enumerate existing edges and open endpoints from `
     Gsub`, then use historical motifs to extend
     candidate paths.
2. Introduce a hidden intermediate node only if it maps
     to a canonical type in P\Omega and is supported by
     a historical motif or evidence summary.
3. Do not skip intermediate nodes that are necessary
     for causal plausibility; if the chain requires a
     missing bridge that cannot be justified, drop the
     candidate.
4. Preserve hourly normalized temporal cues together
     with the spatial and intensity cues required by the
      external deterministic symbolic validator to
     execute `Ctemp`, `Ccausal`, and `Cintensity` on
     every edge after generation.
5. In the current prompt deployment, `located_in` and `
     reported_by` remain attribute-level bindings and
     must not be generated as standalone edges.
6. Return a JSON array only, with no explanatory prose,
     and do not perform algebraic or graph-consistency
     checking inside the model.
</reasoning_procedure>

<inference_policy>
- If a node is directly supported by evidence, mark its
     `status` as `observed`; if it is completed from
     motif or context, mark it as `inferred`.
- Use `resolved` only for chains or nodes that have
     already been confirmed by the feedback/update
     module.
- If no credible candidate chain exists, return `[]`.
- Do not emit schema-external node types, relation
     types, or implausible cross-domain jumps.
</inference_policy>

<symbolic_validation_reference>
The following checks are executed after generation by
     an external deterministic symbolic validator; the
     runtime model is not responsible for filling `
     screening_result` or `violation_tags`.
1. Every edge must remain consistent with the node and
     relation types in P\Omega.
2. The validator checks whether the chain violates `
     Ctemp`, `Ccausal`, or `Cintensity`, and back-
     fills the screening result at edge and chain level.
3. If the path forms a cycle, reverse-temporal
     propagation, or implausible cross-region
     propagation, the validator rejects the chain.
4. Only candidates that pass consistency checking may
     proceed to the write-back stage.
</symbolic_validation_reference>

<output_contract>
Return a JSON array only. Each chain object must
     satisfy:
{
 "chain_id": "...",
 "nodes": [
  {"event_id": "...", "label": "...", "entity_type":
     "...", "time_start": "...", "time_end": null, "
     location_name": "...", "location_level": "...", "
     intensity_level": "...", "status": "observed|
     inferred|resolved"}
 ],
 "edges": [
  {"source": "...", "relation": "...", "target": "..."}
 ],
 "evidence_span": ["..."],
 "confidence": 0.0,
 "status": "observed|inferred|resolved",
 "reason": "...",
 "notes": ""
}
Formatting constraints:
- all `source` and `target` values must reference an `
     event_id` already listed in `nodes`
- if no evidence span exists, use `evidence_span=[]`
```

```
- do not add extra keys
- organize nodes in temporal order when comparable;
    otherwise use causal order
Post-processing note:
- the object above is the candidate schema
- the external symbolic validator appends ‘
    screening_result‘ and ‘violation_tags‘ at edge
    and chain level after generation
</output_contract>
```

Appendix B. Data and code availability

To balance academic transparency with necessary confidentiality and security protocols, this study establishes a bounded reproducibility framework. We explicitly define the operational boundaries of the shared artifacts below.

B.1. Public derived KG artifacts

Owing to institutional confidentiality agreements, third-party redistribution restrictions, and the critical risk that the combination of explicit temporal, spatial, infrastructure damage, and local response vectors could expose sensitive operational intelligence regarding disaster response, the raw multi-source textual corpus cannot be publicly distributed.

To ensure comprehensive auditability despite the constraints on source data, we provide the full, de-identified derived knowledge-graph artifacts constructed during training. The released dataset encompasses 99,026 de-identified nodes and 127,940 directed edges, partitioned into 13,208 connected components (with the largest component spanning 731 nodes alongside 2665 singleton components). This graph asset includes complete inventories of relations, classifications of evidence types, topological summaries at the component level, validation flags for constraints, and identifiers for training and validation splits.

B.2. Code, configuration, and evaluation utilities

The complete programmatic execution pipeline is open for inspection. The released code comprises:

- The core implementation of the *NeuroSymbolic-Chain* framework.
- The definitive, production-ready `YAML` configuration templates that map all optimized neural and symbolic hyperparameters (matching Appendix A.1).
- The machine-readable `OWL/TTL` serialization of the P_{Ω} disaster ontology (matching Appendix A.2).
- The structured prompt templates utilized across the extraction and feedback loops (matching Appendix A.3).
- The programmatic evaluation scripts, cross-paradigm evaluation contexts for operations (including integration hooks for SWMM), and utilities for the graph-level audit of factual consistency.

B.3. Reproducibility boundary

The public artifacts are hosted across two dedicated, version-controlled repositories:

- **Code Framework Repository:** Code repository
- **Derived KG Dataset Repository:** Derived KG repository

These resources support KG-level structural audits, graph-statistics verification, logical-constraint validation, hyperparameter inspection, and evaluation-pipeline execution. However, they do not support reconstruction of the restricted raw corpus or end-to-end text extraction from confidential source materials.

Appendix C. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.knosys.2026.116650.

Data availability

Data will be made available on request.

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